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(54) **AUTOMATIC RETRACTABLE SCREEN DEVICE FOR STORM DRAINS HAVING CURBSIDE OPENINGS**

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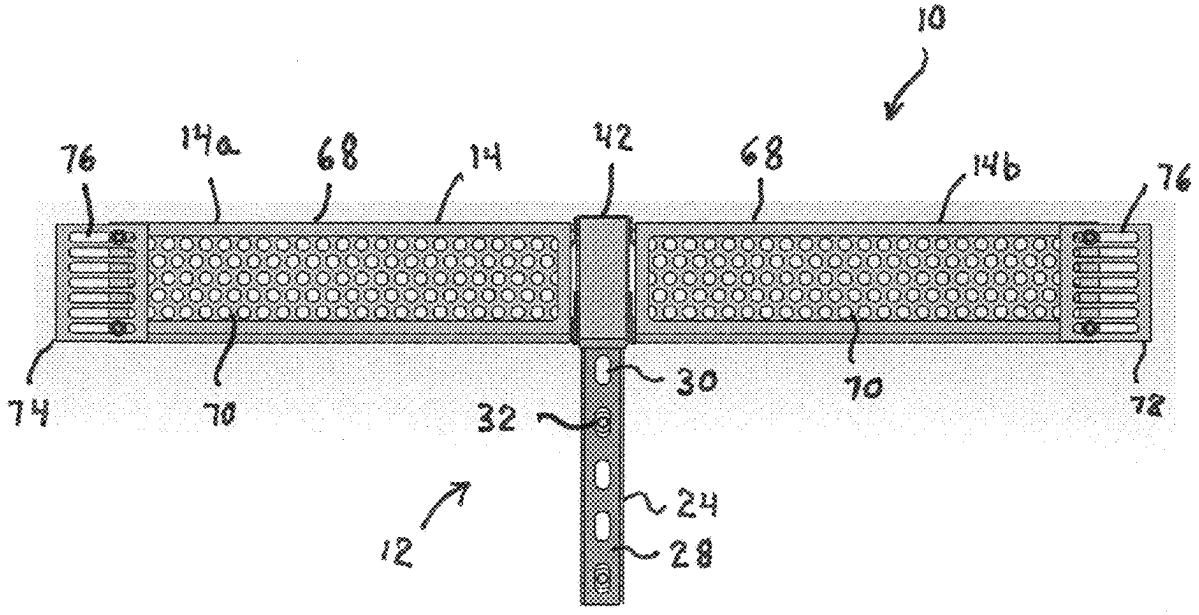
C02F 103/00	(2006.01)

(52) **U.S. Cl.**

CPC ***E03F 5/14*** (2013.01); ***B01D 29/05*** (2013.01); ***B01D 35/02*** (2013.01); ***C02F 1/001*** (2013.01); *B01D 2201/4069* (2013.01); *C02F 2103/001* (2013.01)

(57) **ABSTRACT**

An automatic retractable screen device is mountable within the interior cavity of a storm vault and adjacent to a vault opening formed in the side wall through which stormwater flows and into the interior cavity. The automatic retractable screen device includes a mounting assembly for mounting the automatic retractable screen device to the side wall of the storm vault, a pivotable screen positionable to be in alignment with the vault opening of the storm vault, structure affixed to the mounting assembly and the screen and configured to allow the screen to pivot on the mounting assembly between a closed position and an open position, and a magnetic release mechanism operatively coupled to the screen. The magnetic release mechanism holds the screen in the closed position to block debris carried by the stormwater from entering the vault opening until the force of the stormwater exerted on the screen exceeds a predetermined value at which point the screen is released to pivot to the open position.



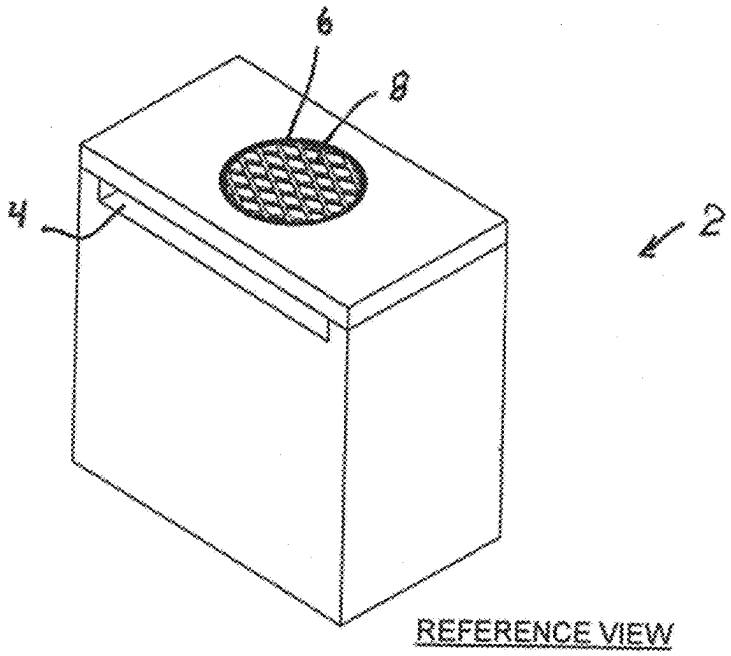

6
8
4
2
REFERENCE VIEW

FIG. 1

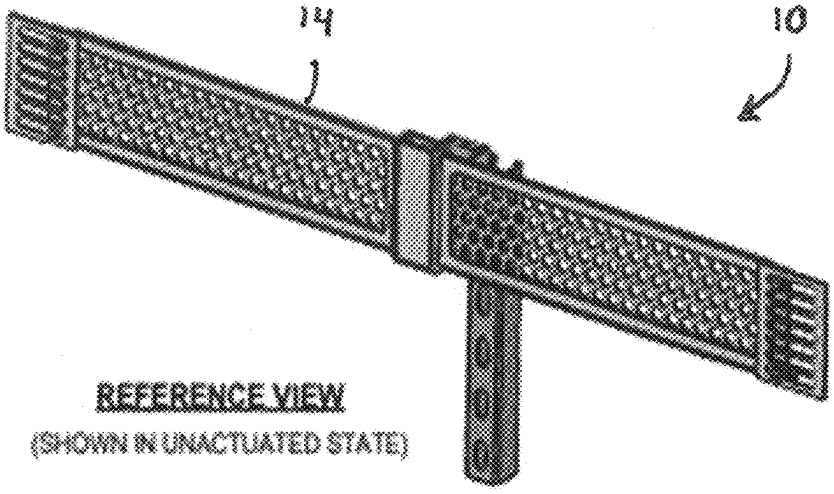

14
10
REFERENCE VIEW
(SHOWN IN UNACTUATED STATE)

FIG. 2

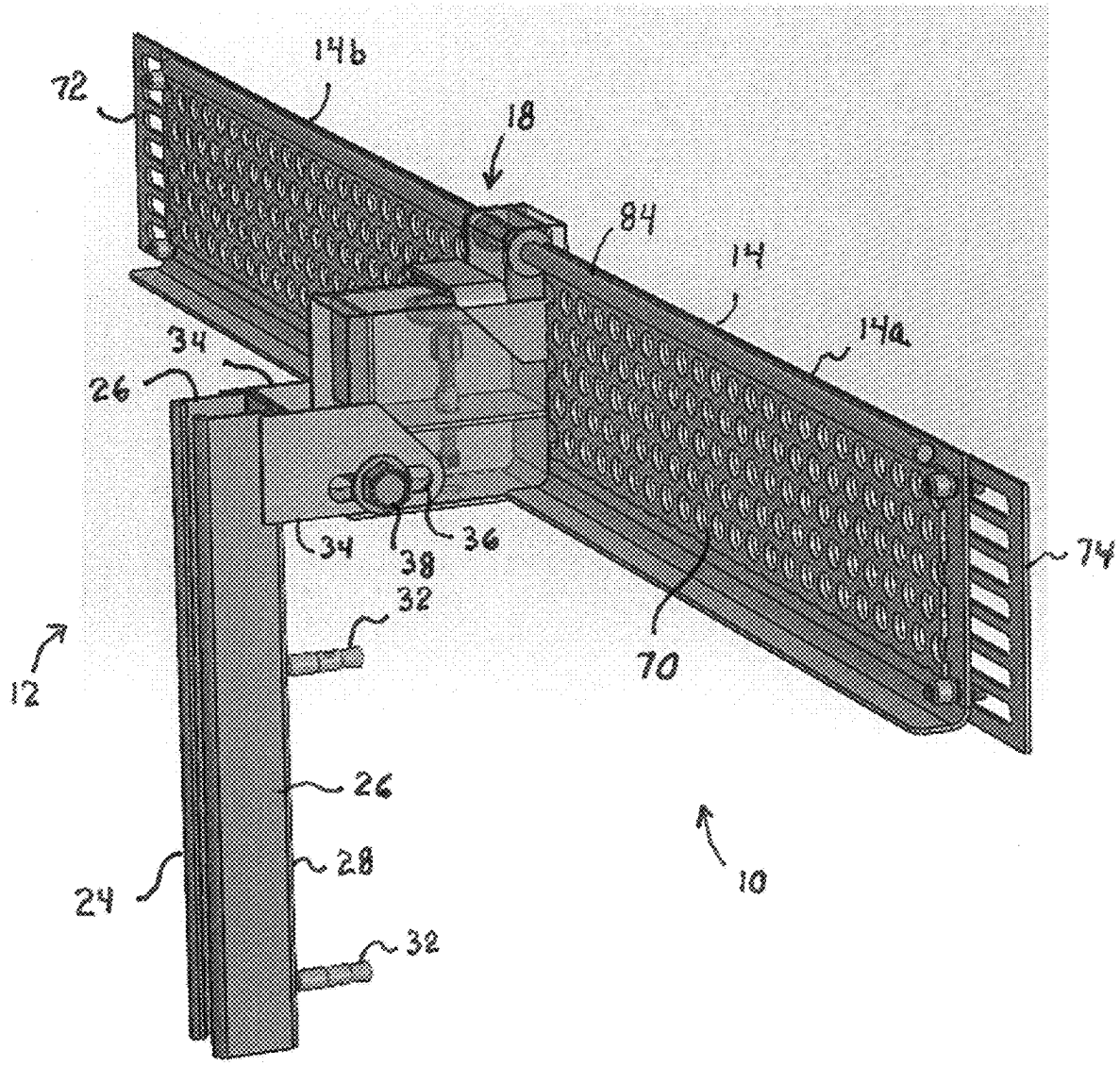


FIG. 3

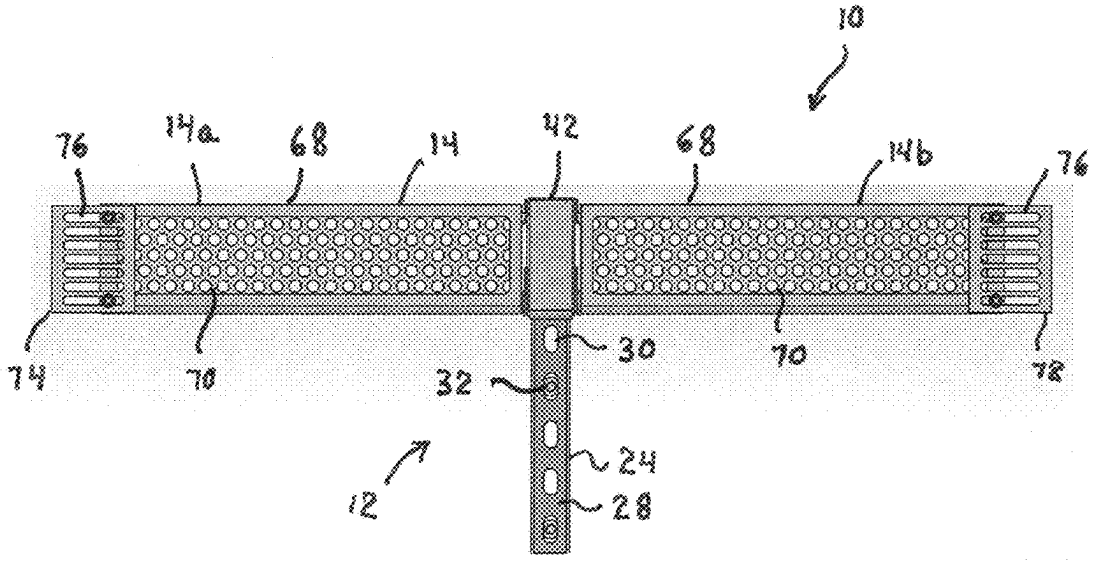

10
76
14a
68
14
42
68
14b
76
74
70
30
70
78
32
24
12
28

FIG. 4

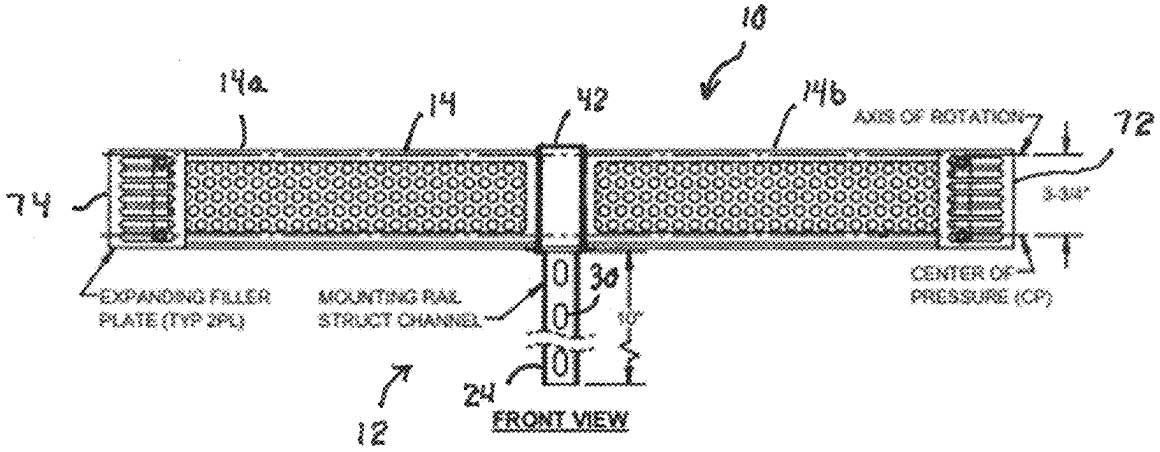

10
14a
14
42
14b
AXIS OF ROTATION
72
74
9.38"
EXPANDING FILLER PLATE (TYP 2PL)
MOUNTING RAIL STRUCT CHANNEL
30
CENTER OF PRESSURE (CP)
24
12
FRONT VIEW

FIG. 5

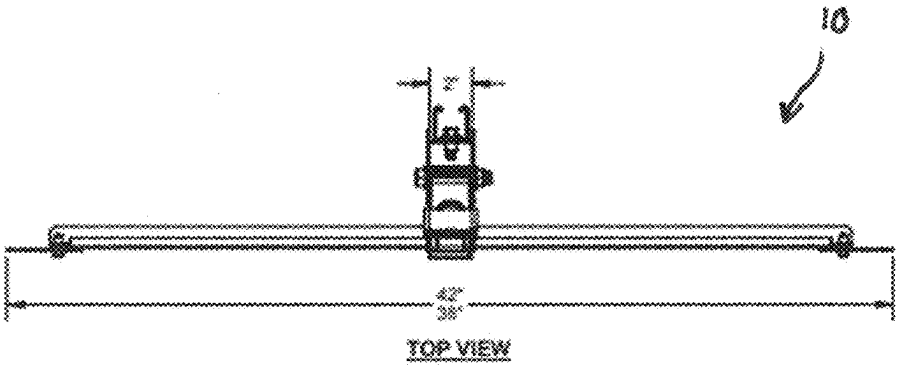


FIG. 6

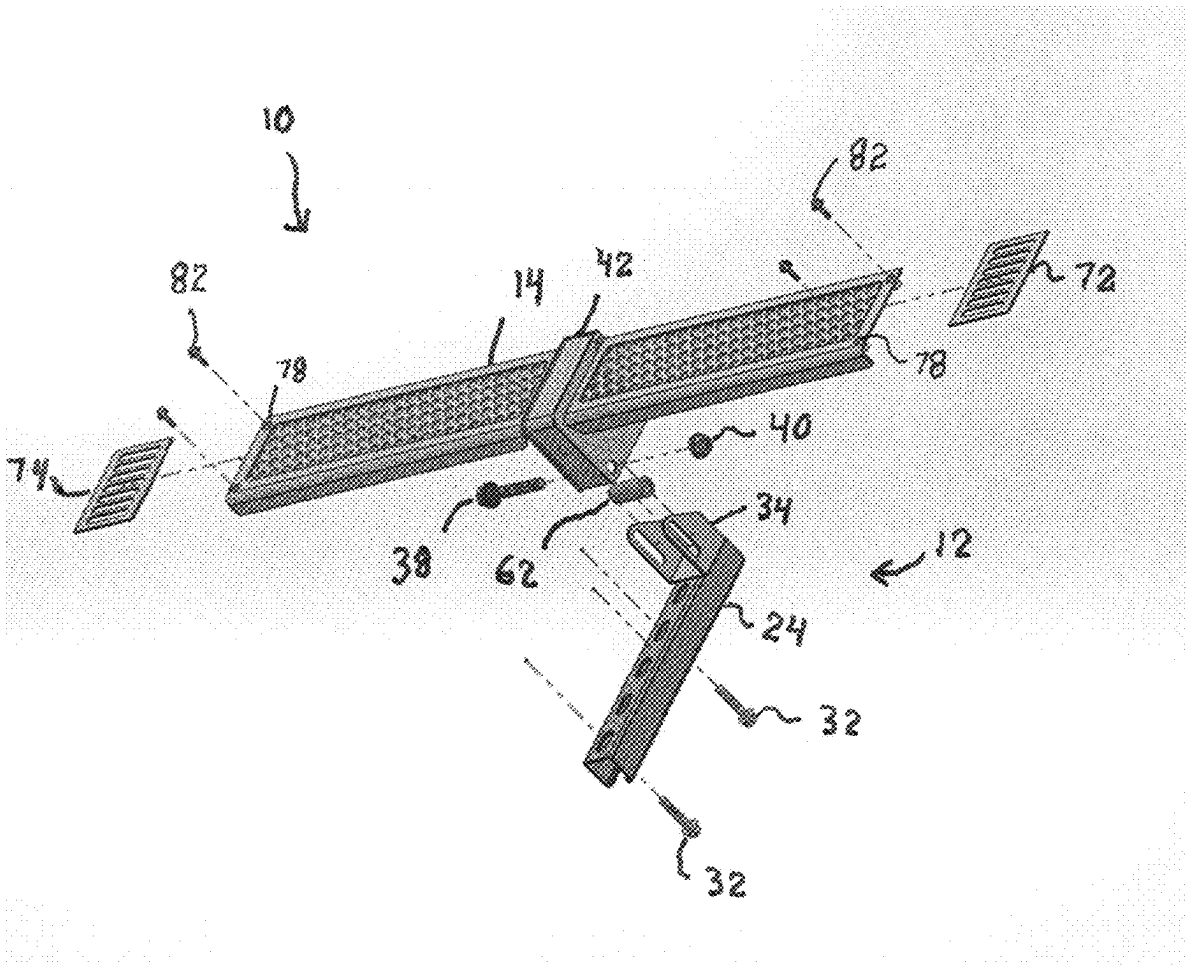


FIG. 7

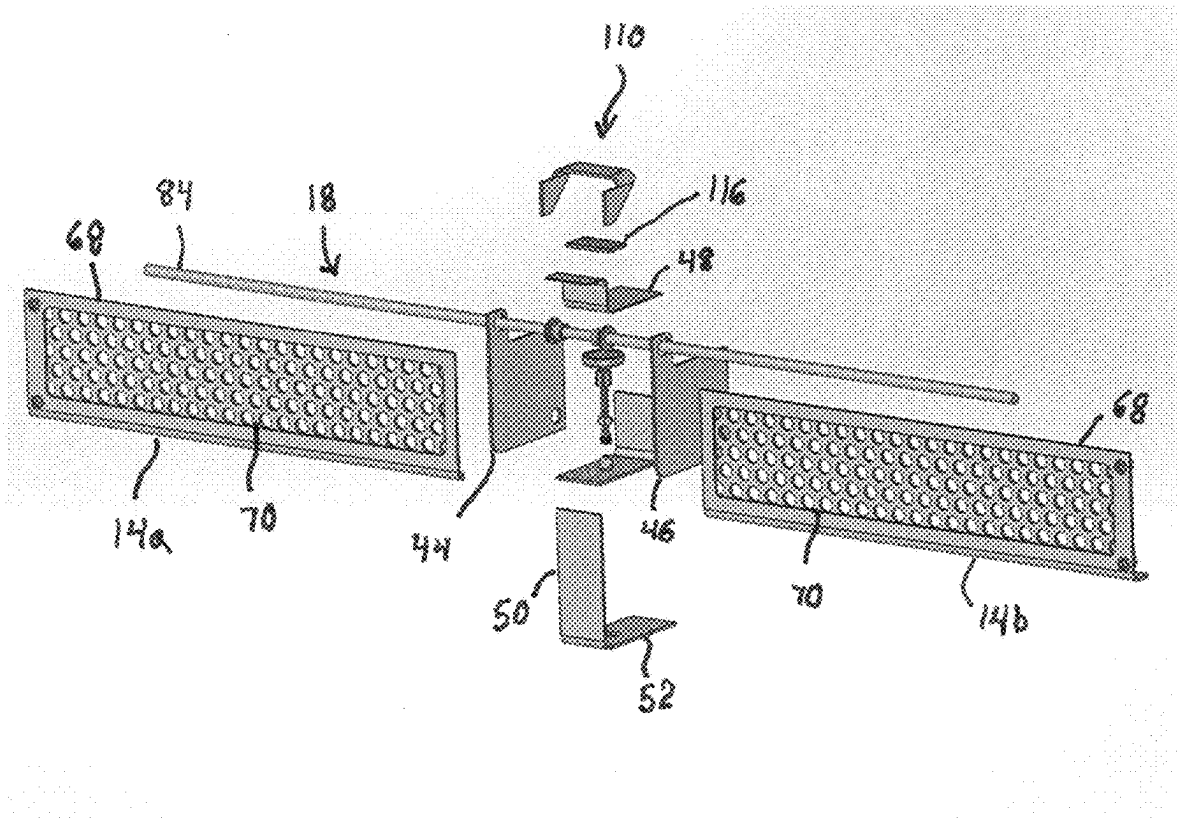


FIG. 8

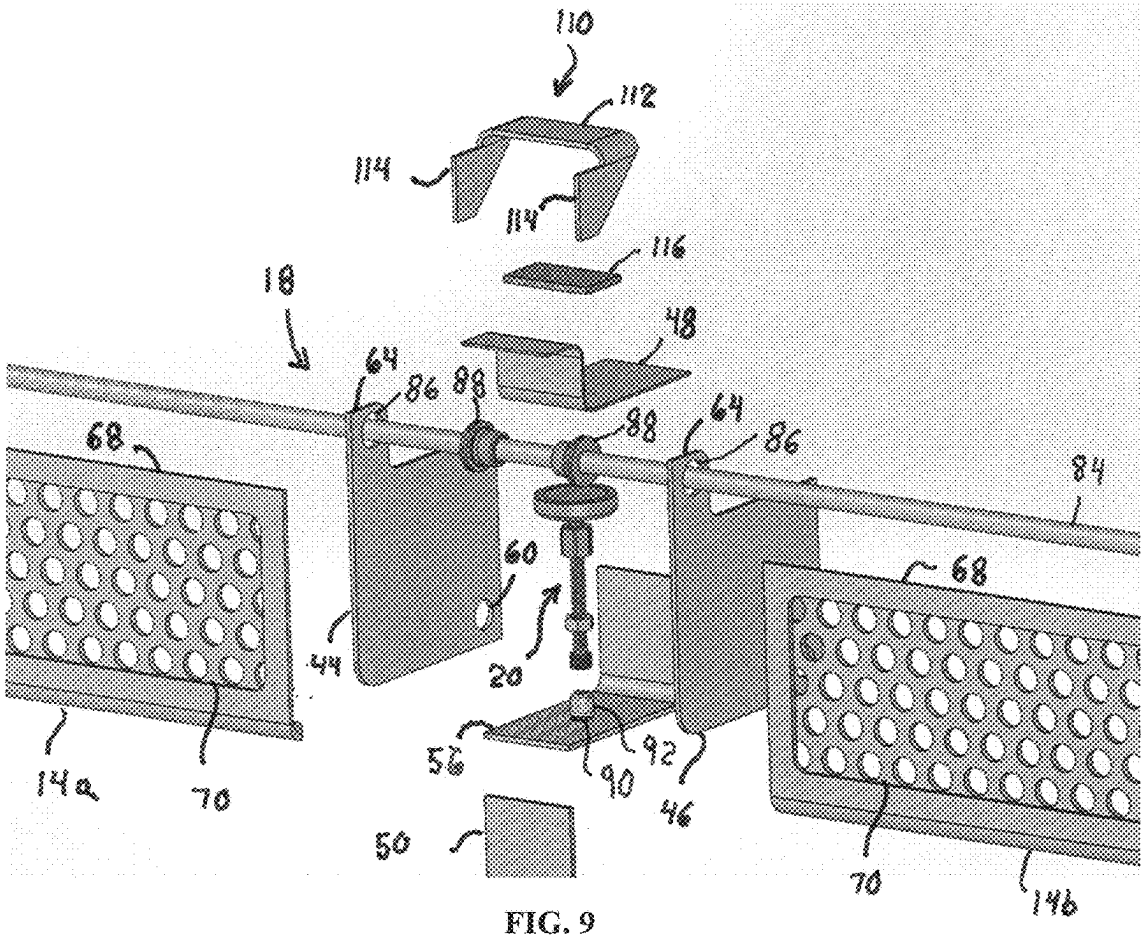

110
112
114
114
116
18
48
64
86
88
88
64
86
68
84
60
68
44
20
56
92
90
46
50
14a
70
70
14b

FIG. 9

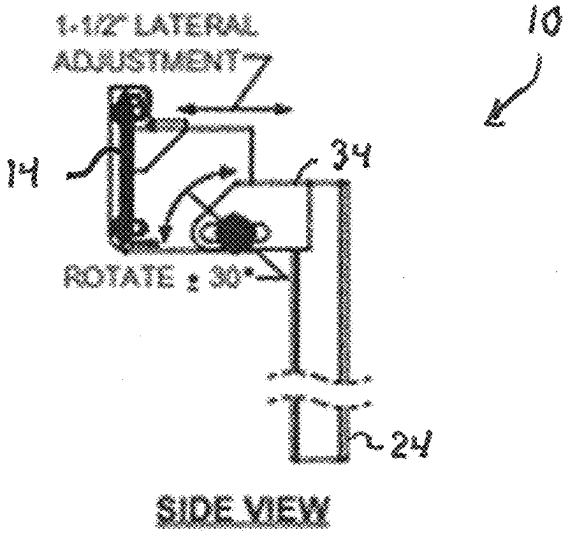


FIG. 10

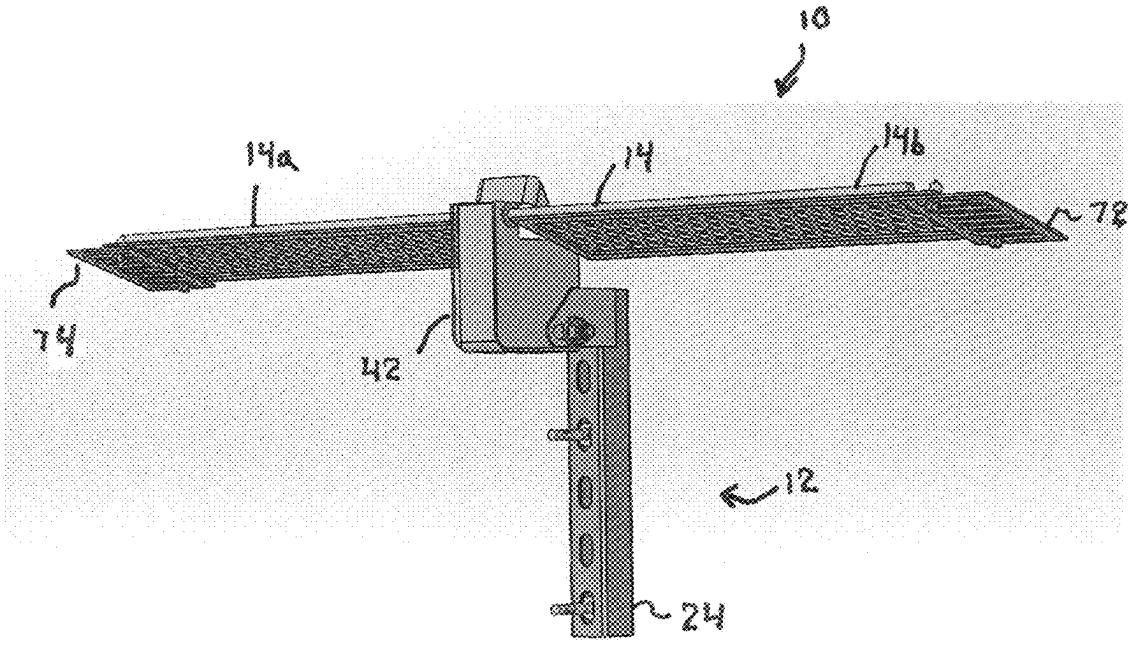


FIG. 11

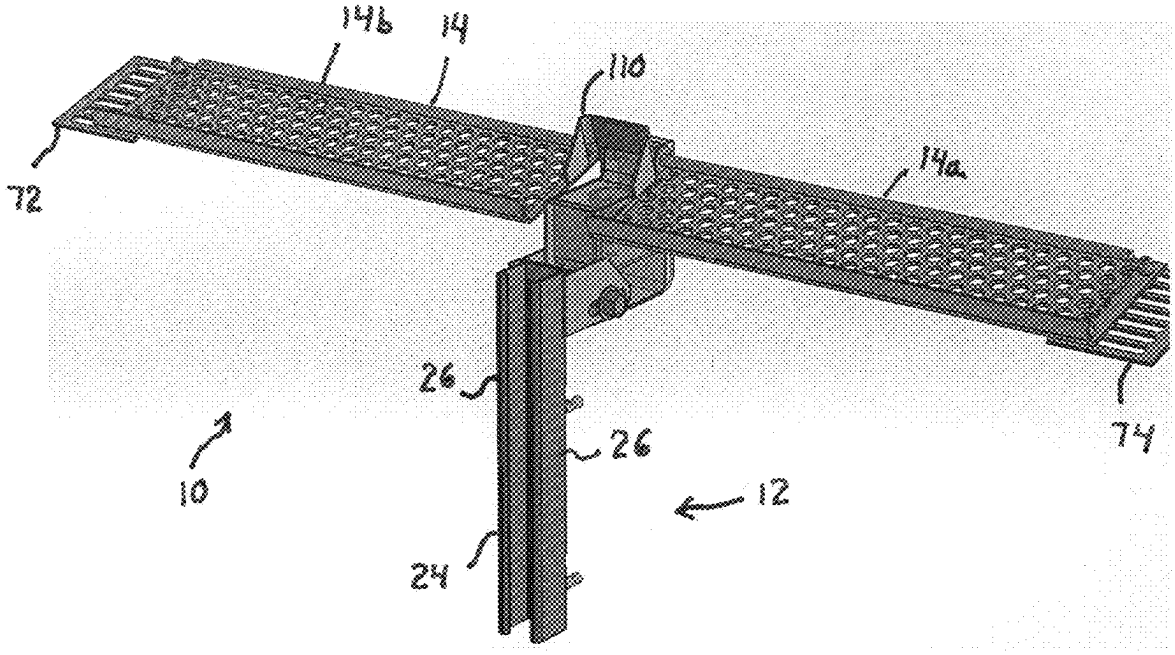


FIG. 12

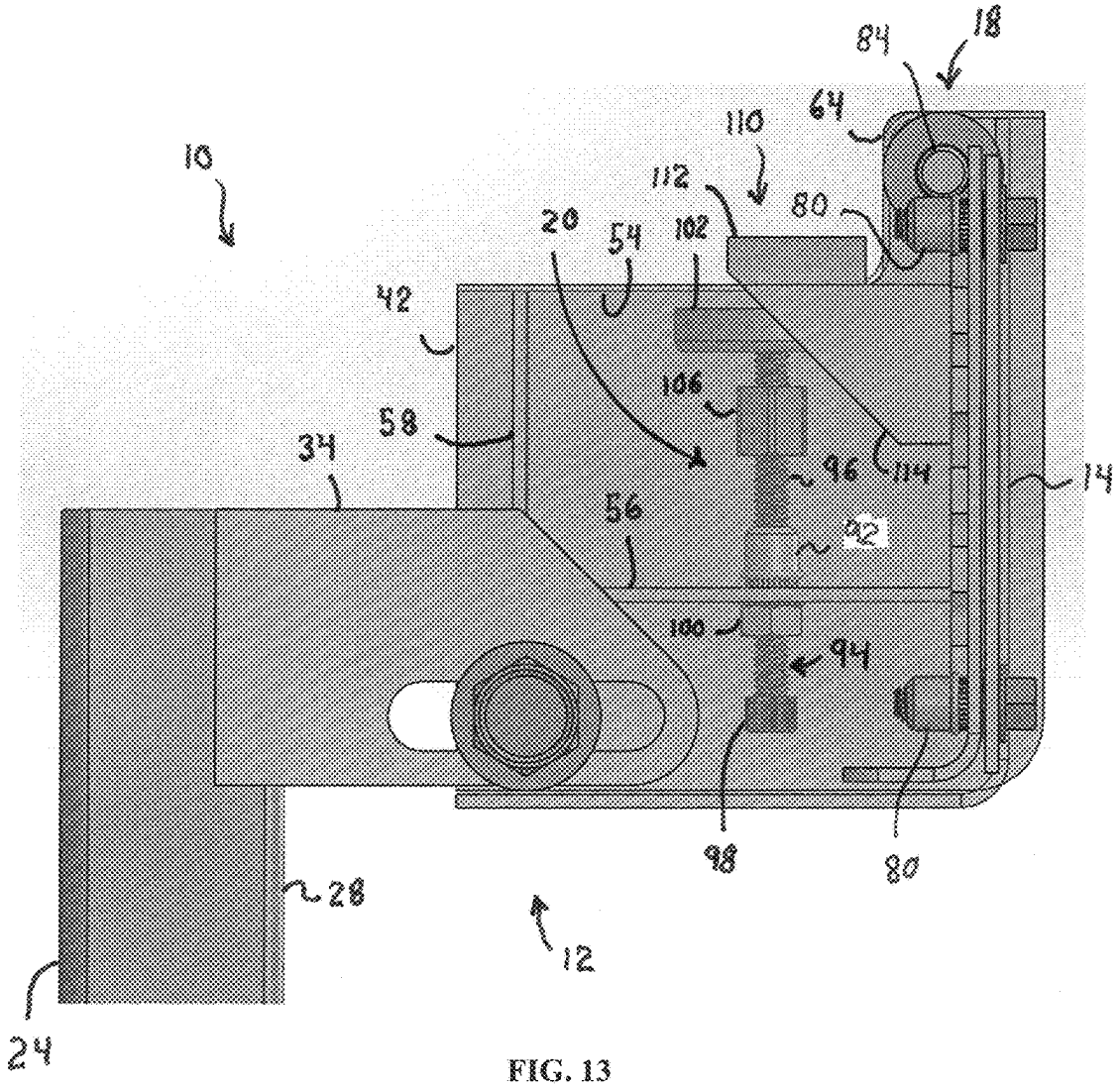

84
18
64
110
10
112
80
20
54
102
42
106
58
34
96
114
14
56
82
100
94
98
80
28
12
24

FIG. 13

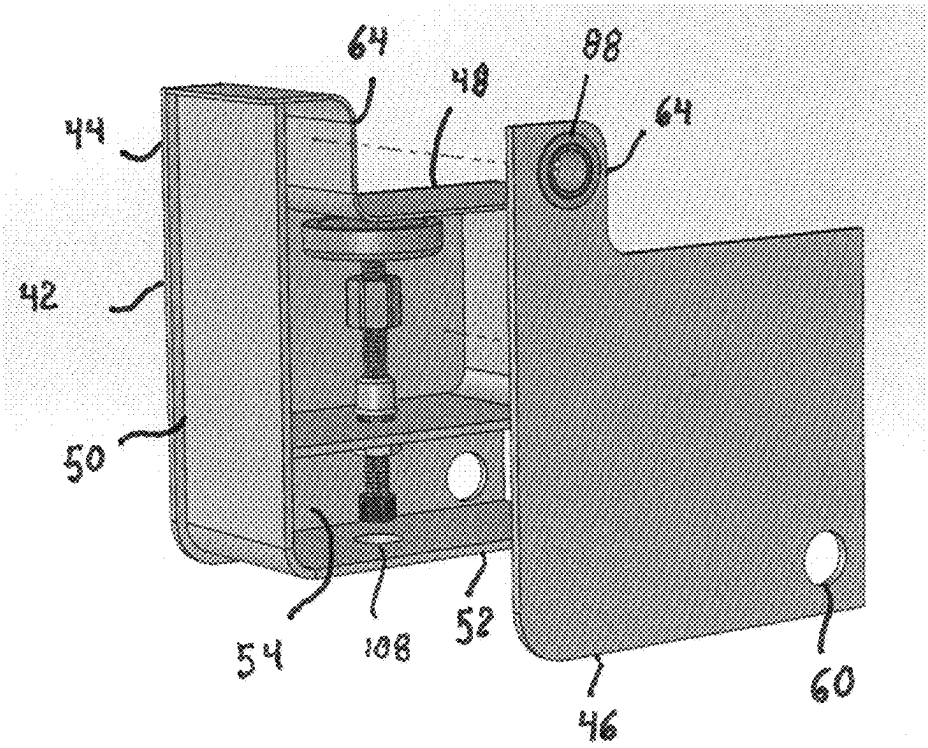


FIG. 14

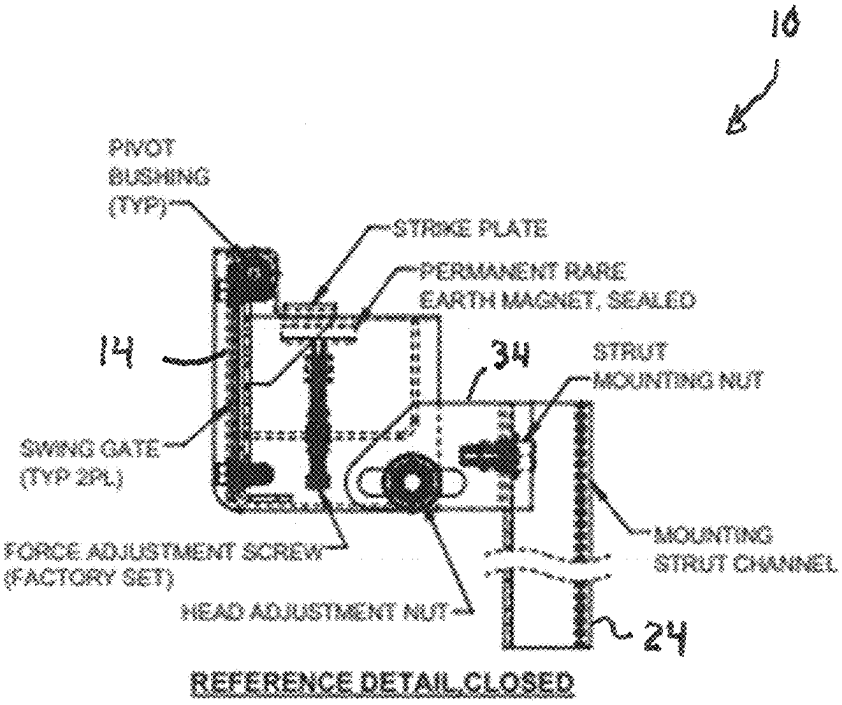

16
PIVOT BUSHING (TYP)
STRIKE PLATE
PERMANENT RARE EARTH MAGNET, SEALED
14
34
STRUT MOUNTING NUT
SWING GATE (TYP 2PL)
FORCE ADJUSTMENT SCREW (FACTORY SET)
HEAD ADJUSTMENT NUT
MOUNTING STRUT CHANNEL
24
REFERENCE DETAIL CLOSED

FIG. 15

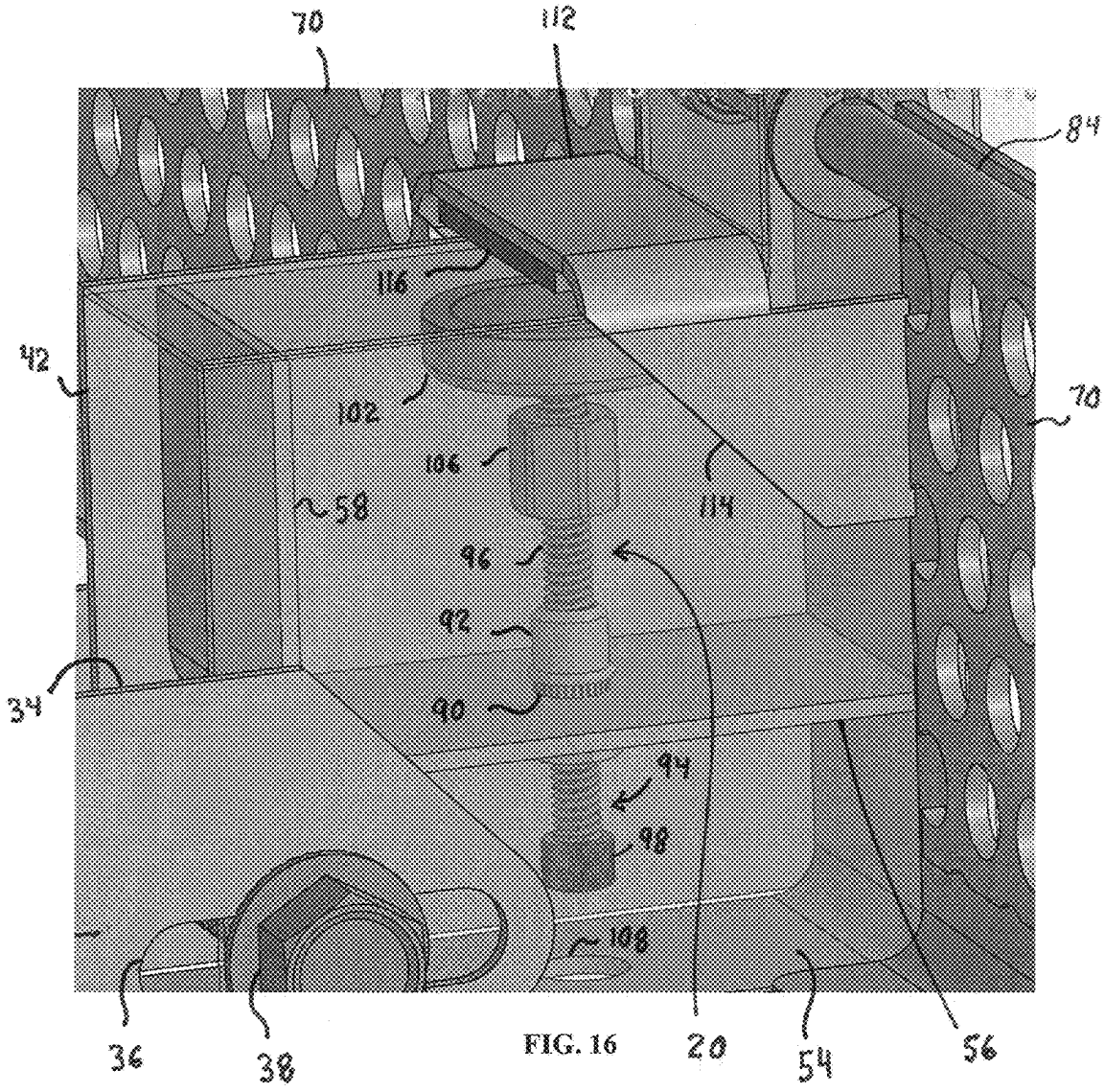

70
112
84
116
42
102
70
106
114
58
96
92
34
90
94
98
108
36
38
20
54
56

FIG. 16

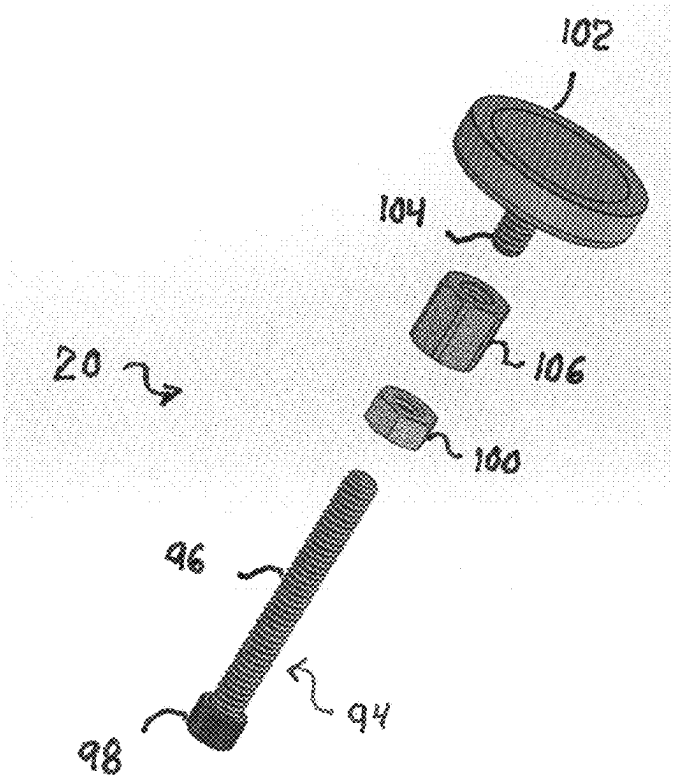


FIG. 17

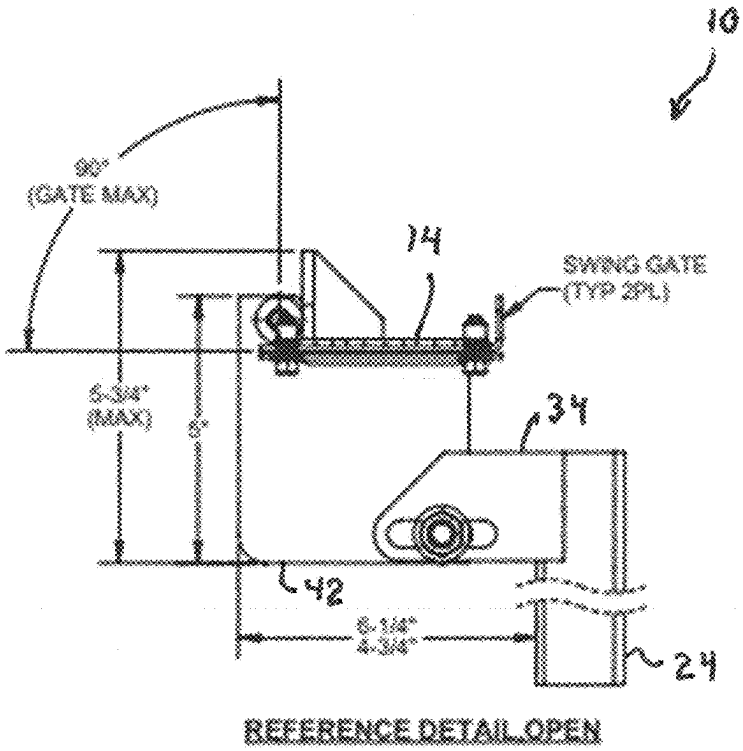

10
90°
(GATE MAX)
14
SWING GATE
(TYP 2PL)
5-3/4"
(MAX)
6"
34
42
6-1/4"
±3/4"
24
REFERENCE DETAIL OPEN

FIG. 18

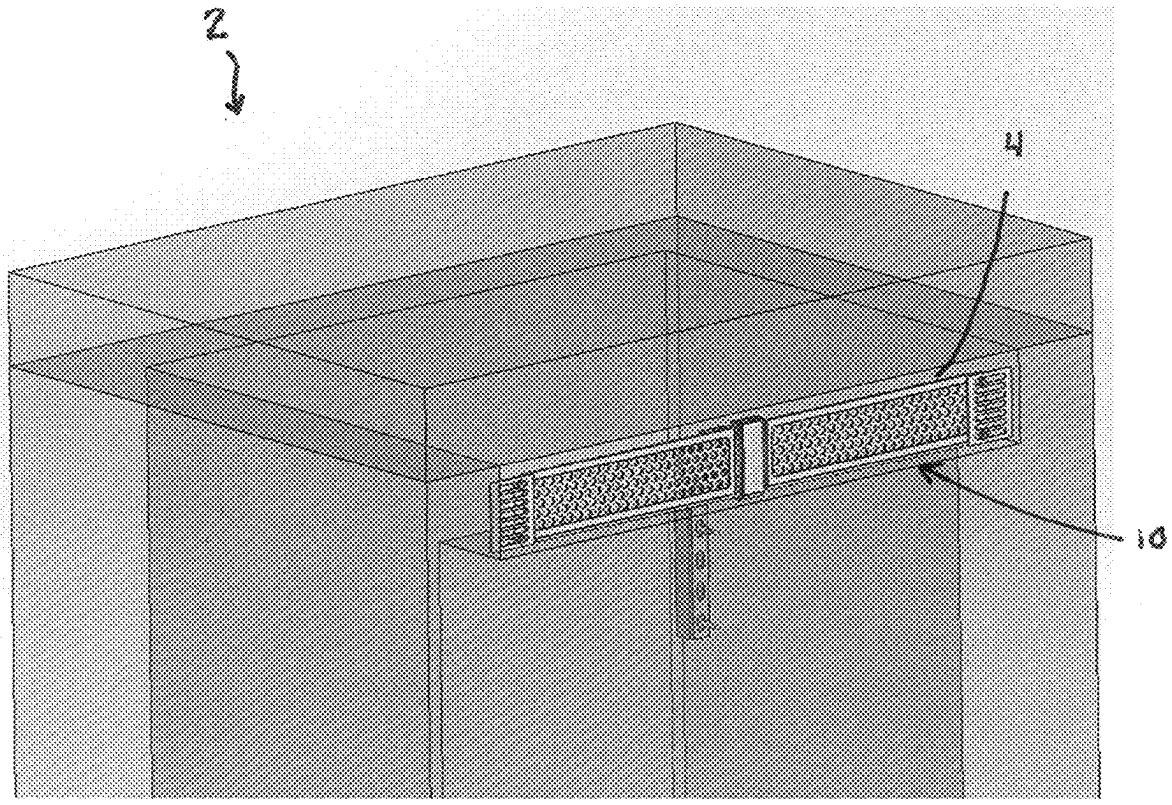


FIG. 19

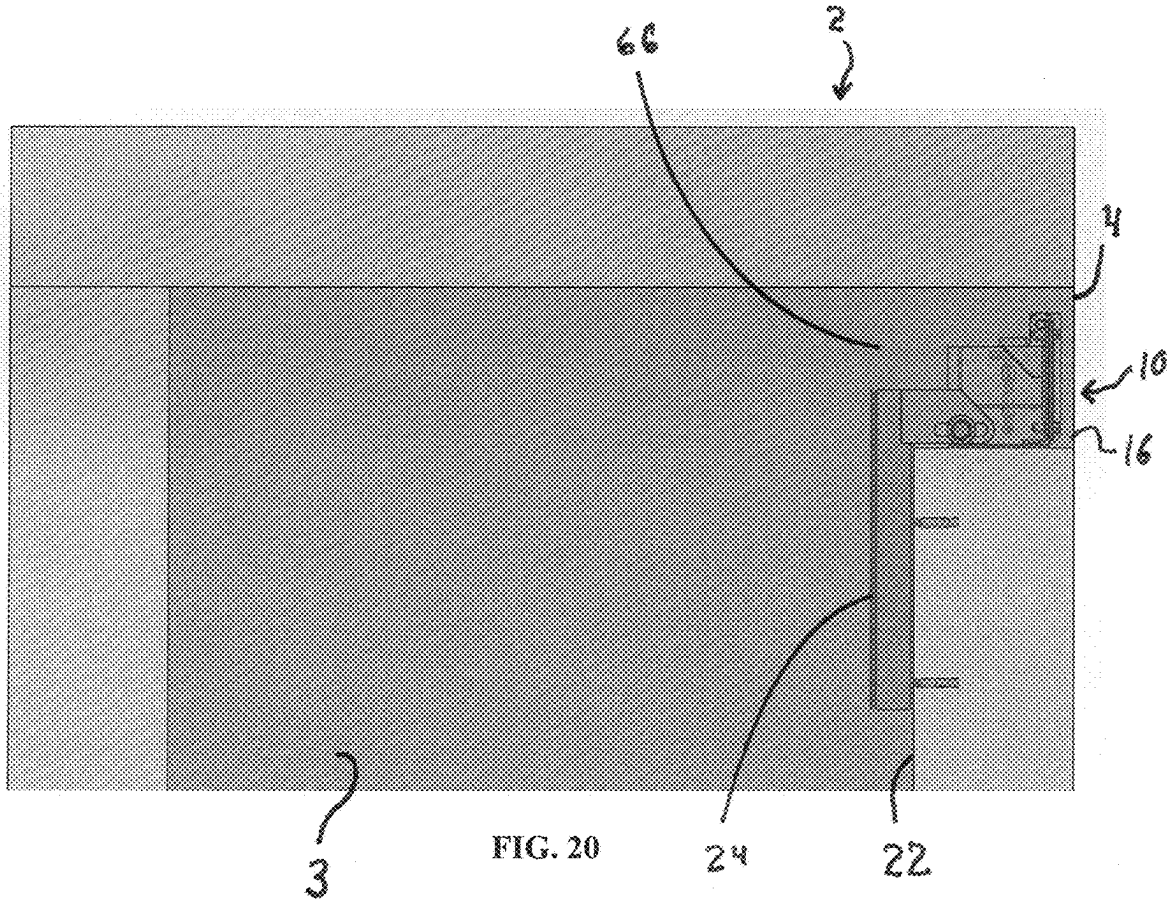


FIG. 20

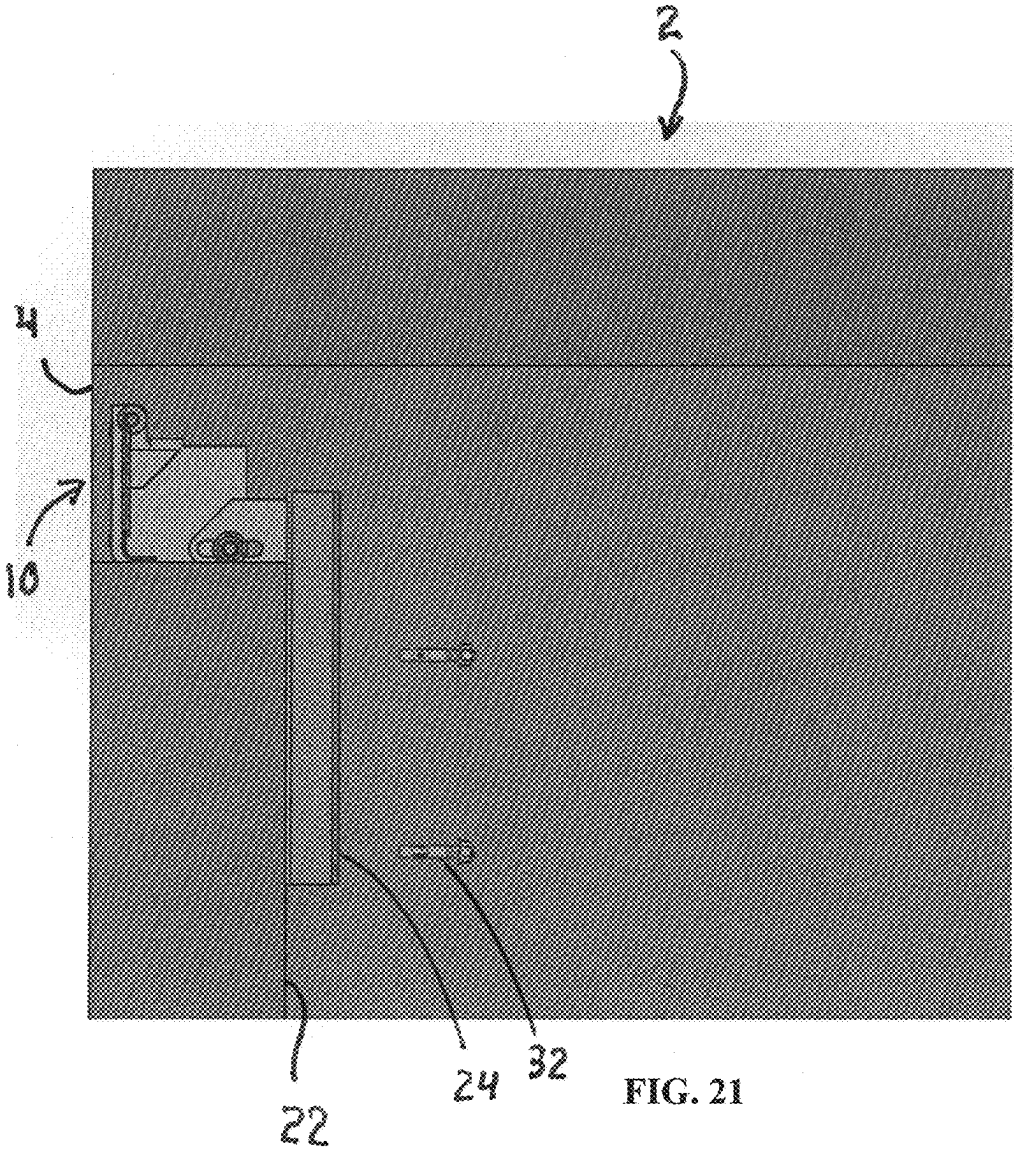


FIG. 21

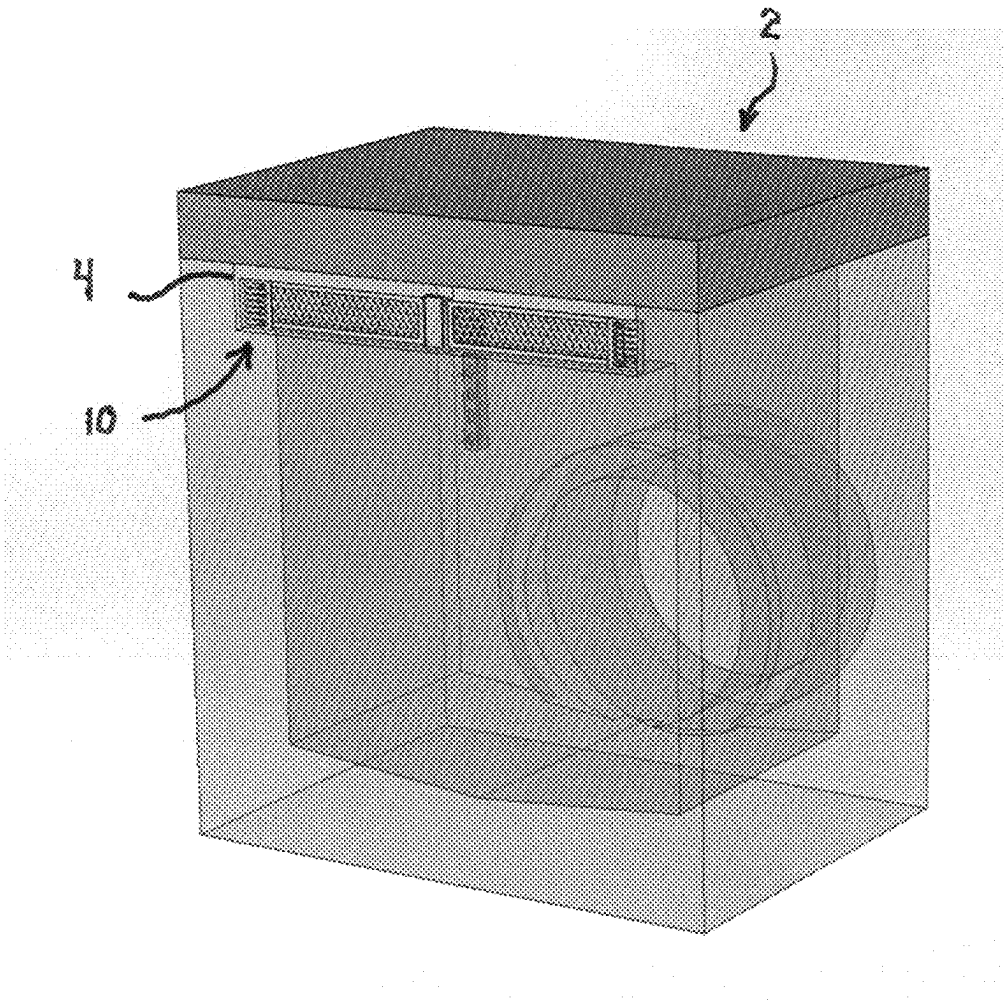


FIG. 22

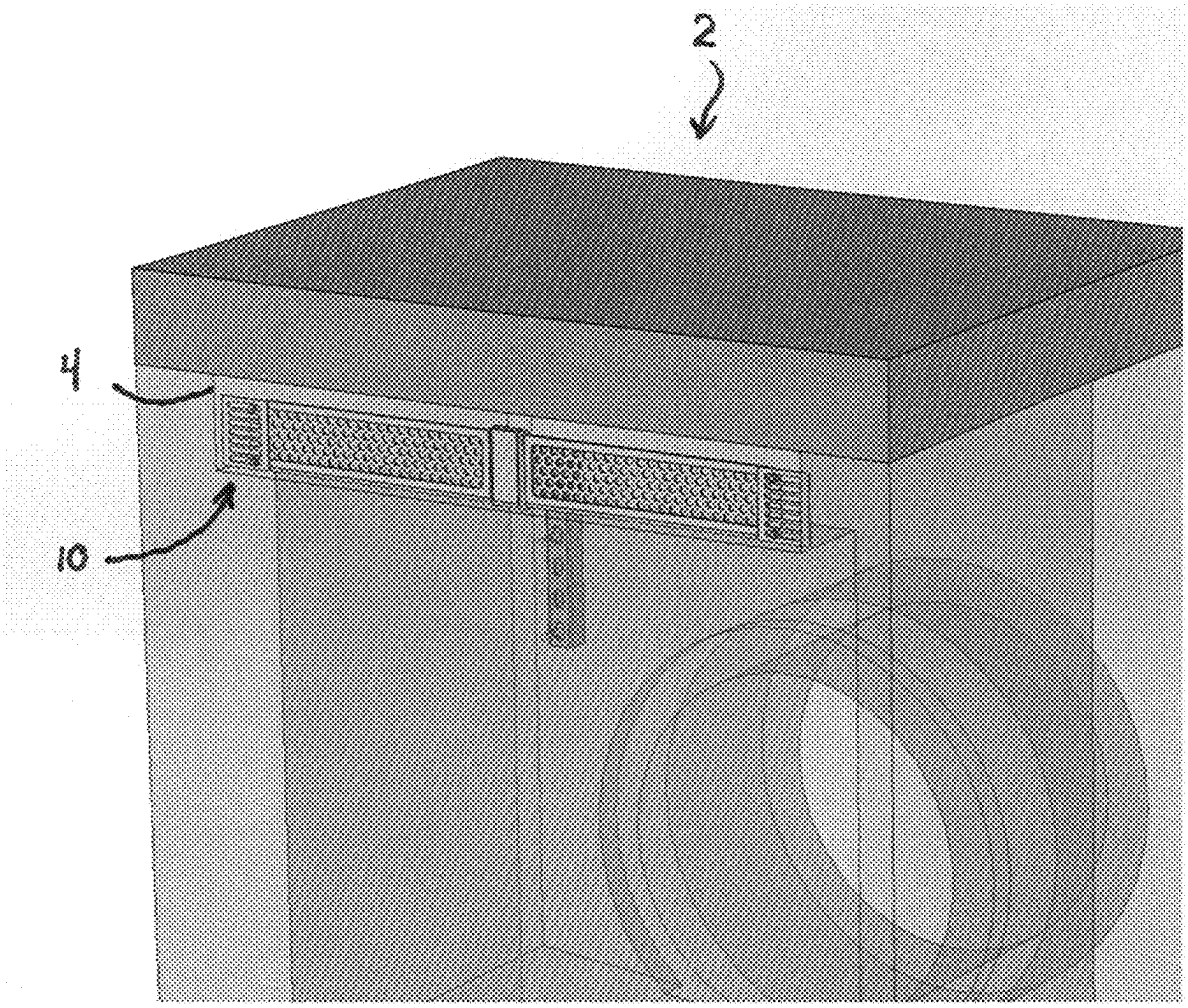


FIG. 23

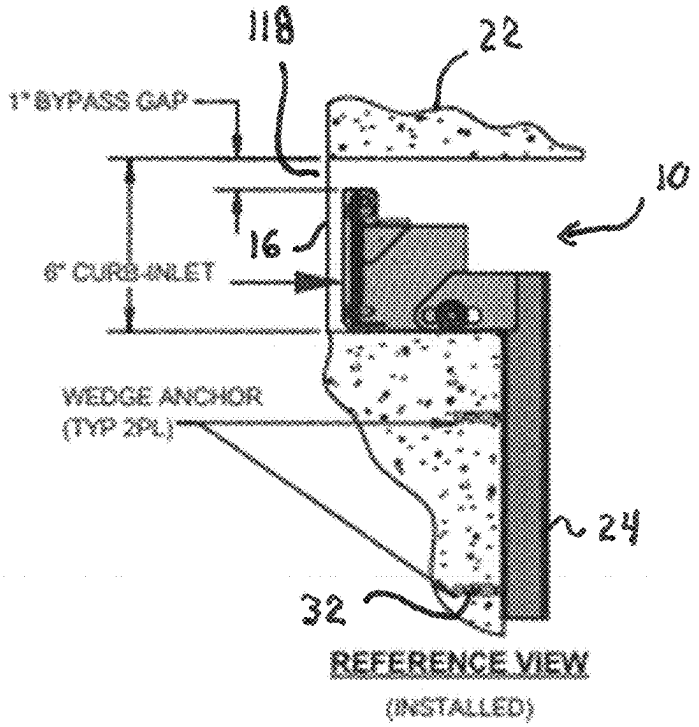

118
22
1" BYPASS GAP
10
16
6" CURB INLET
WEDGE ANCHOR
(TYP 2PL)
24
32
REFERENCE VIEW
(INSTALLED)

FIG. 24

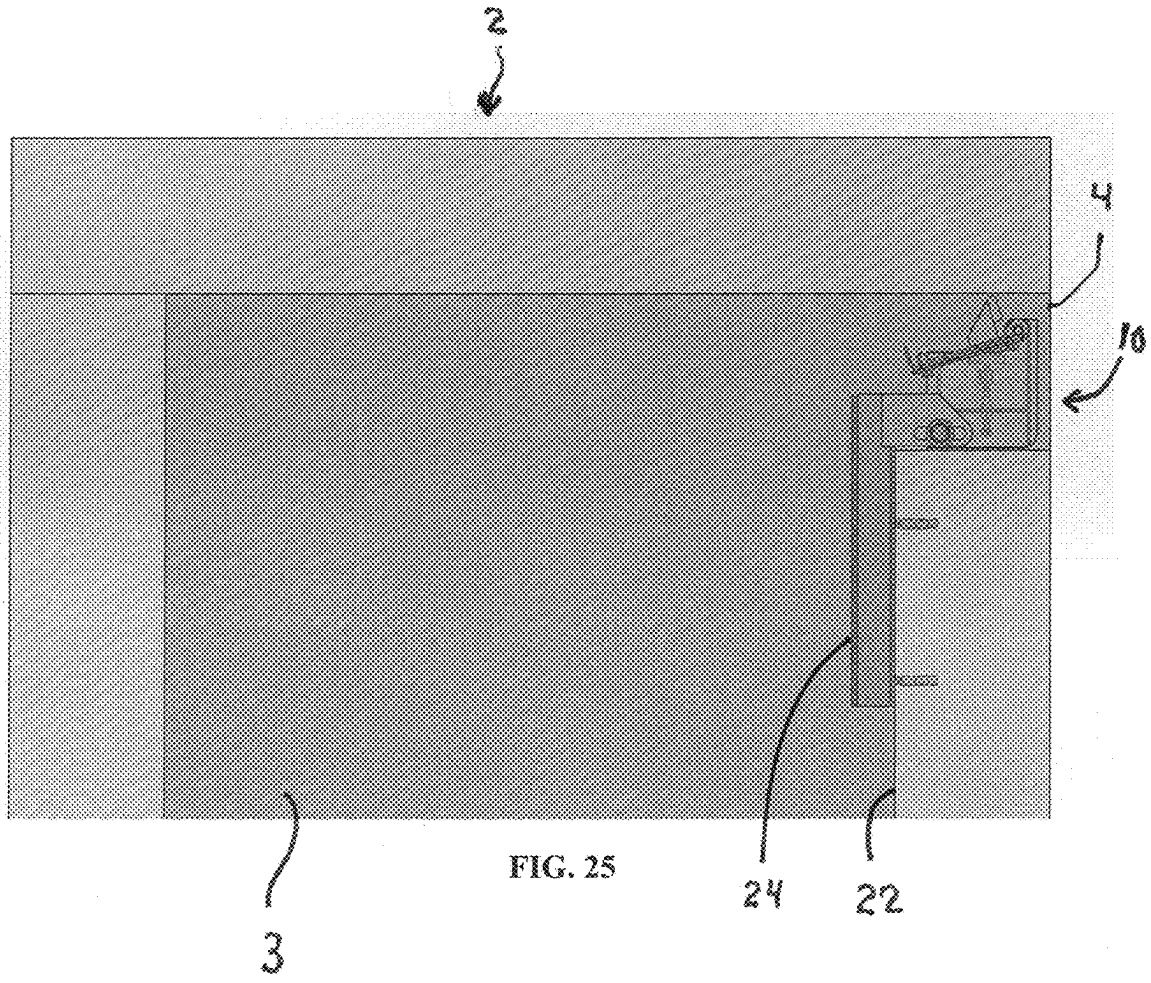


FIG. 25

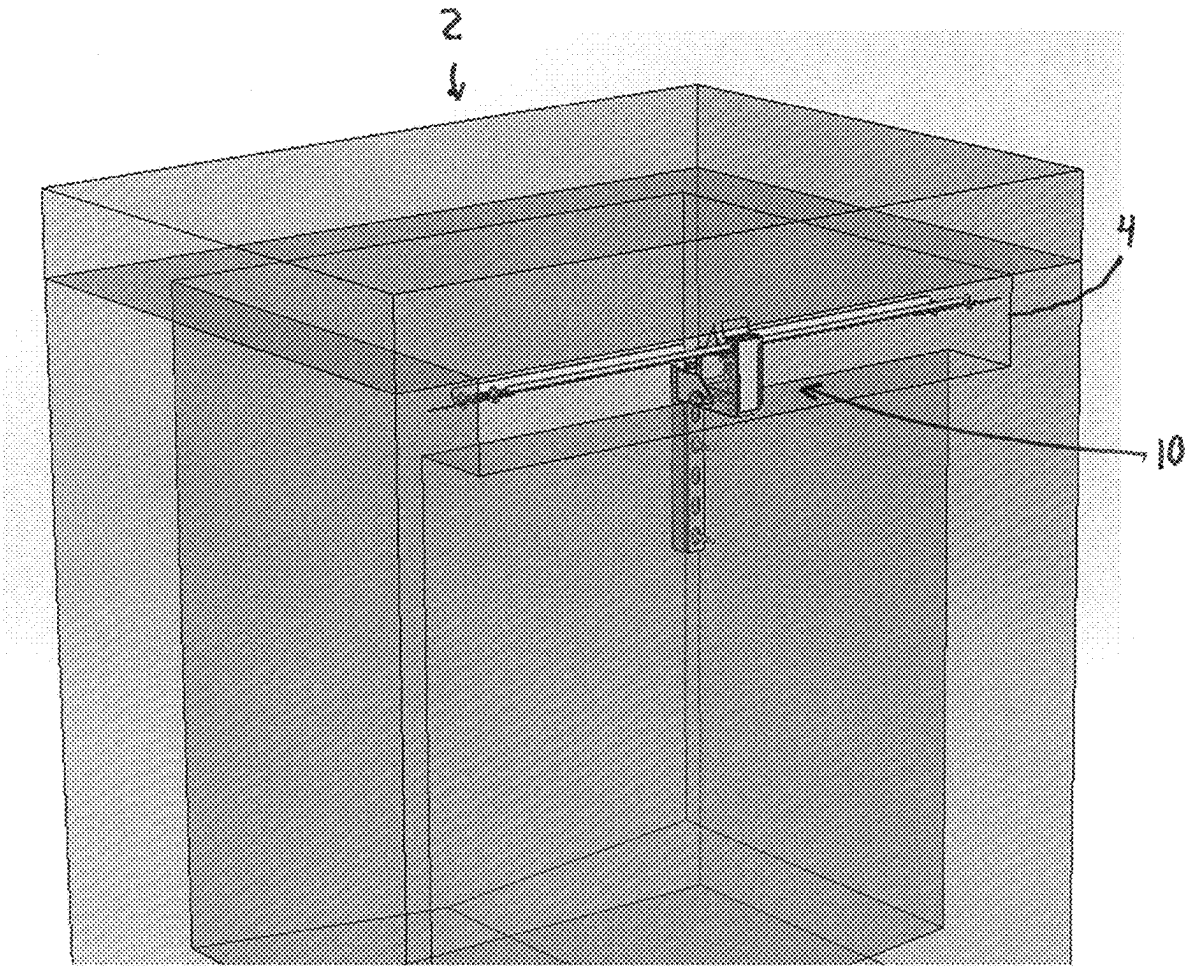


FIG. 26

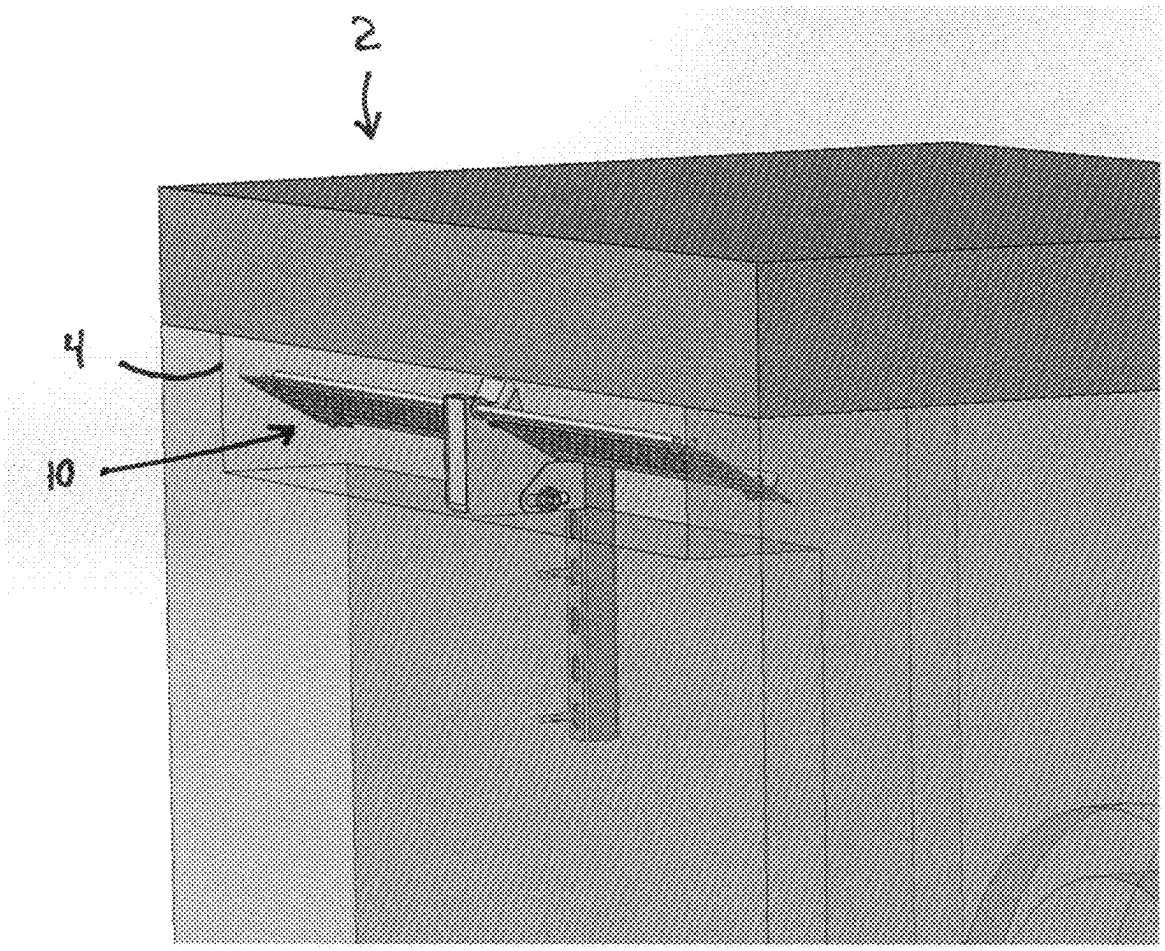


FIG. 27

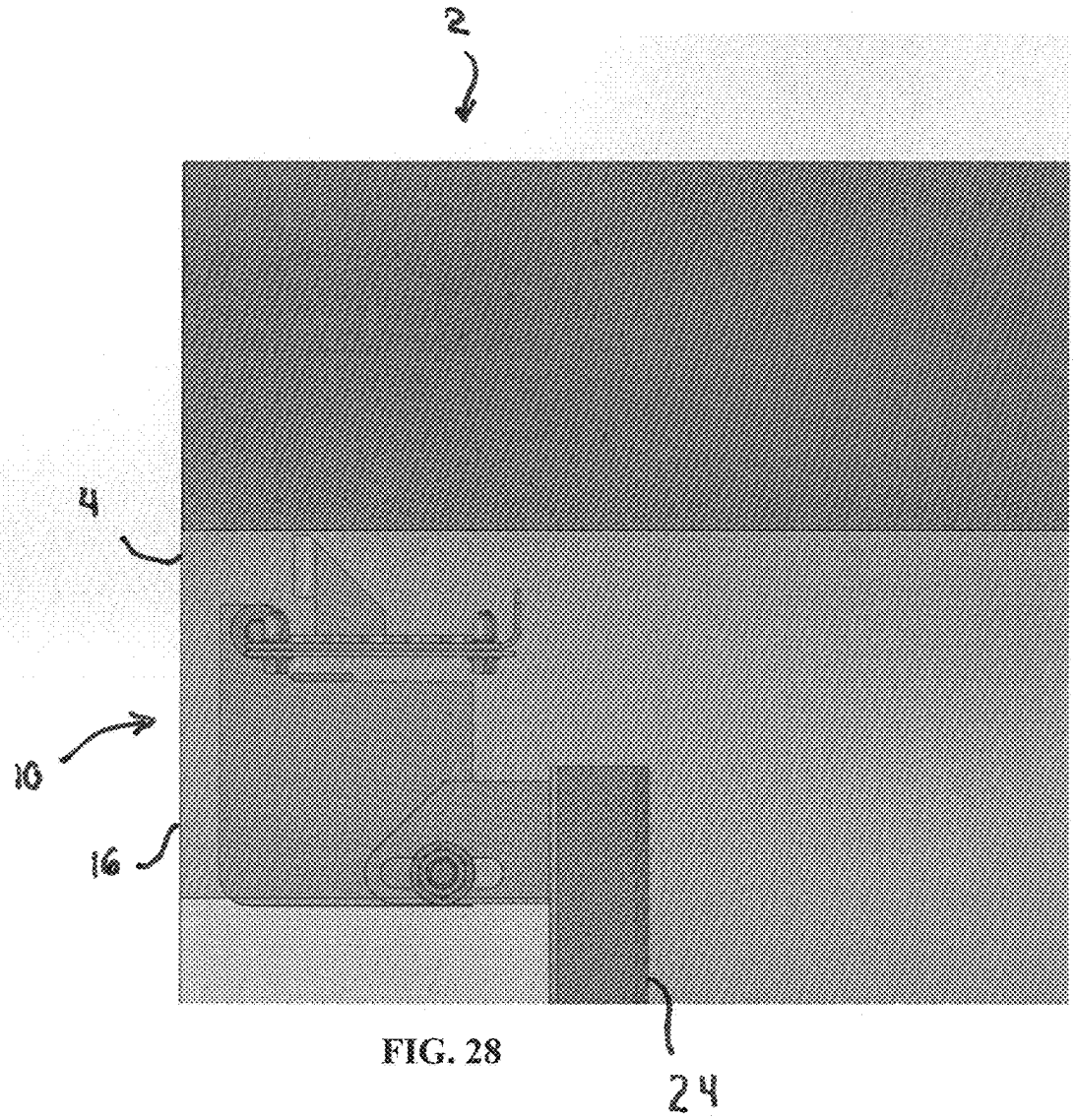


FIG. 28

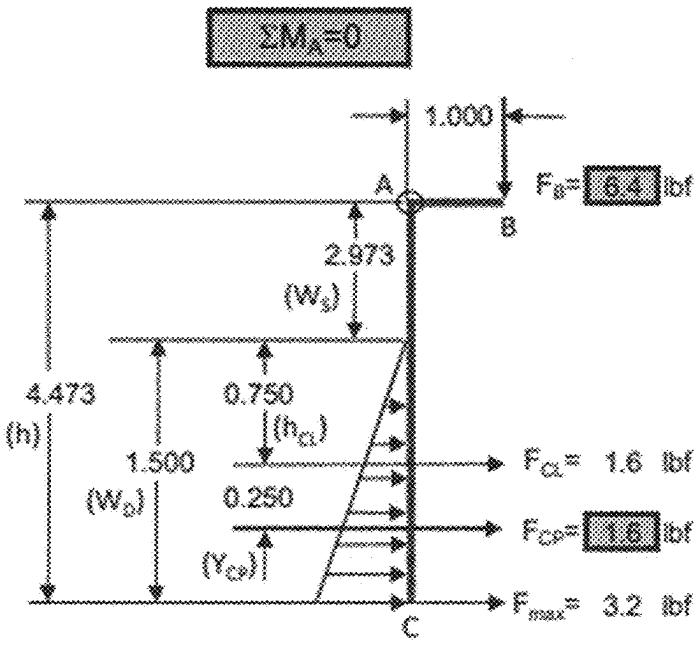

ΣM_A=0
1.000
A
B
F_B= 6.4 lbf
2.973
(W_d)
4.473
(h)
0.750
(h_CL)
1.500
(W_d)
0.250
(Y_CP)
F_CL= 1.6 lbf
F_CP= 1.6 lbf
F_max= 3.2 lbf
C

FIG. 29

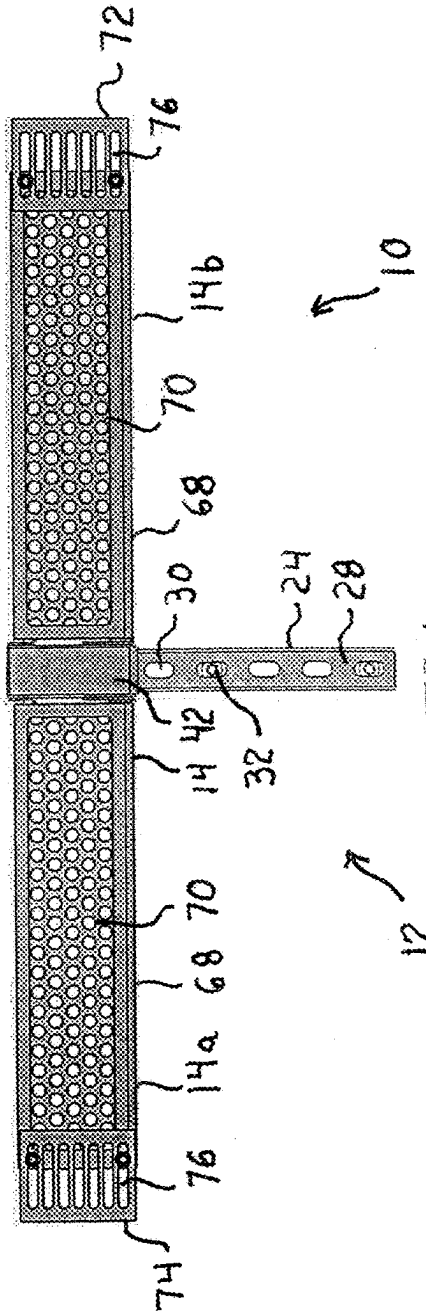

72
76
14b
70
68
30
24
28
10
42
32
14
70
68
14a
76
74
12

FIG. 4

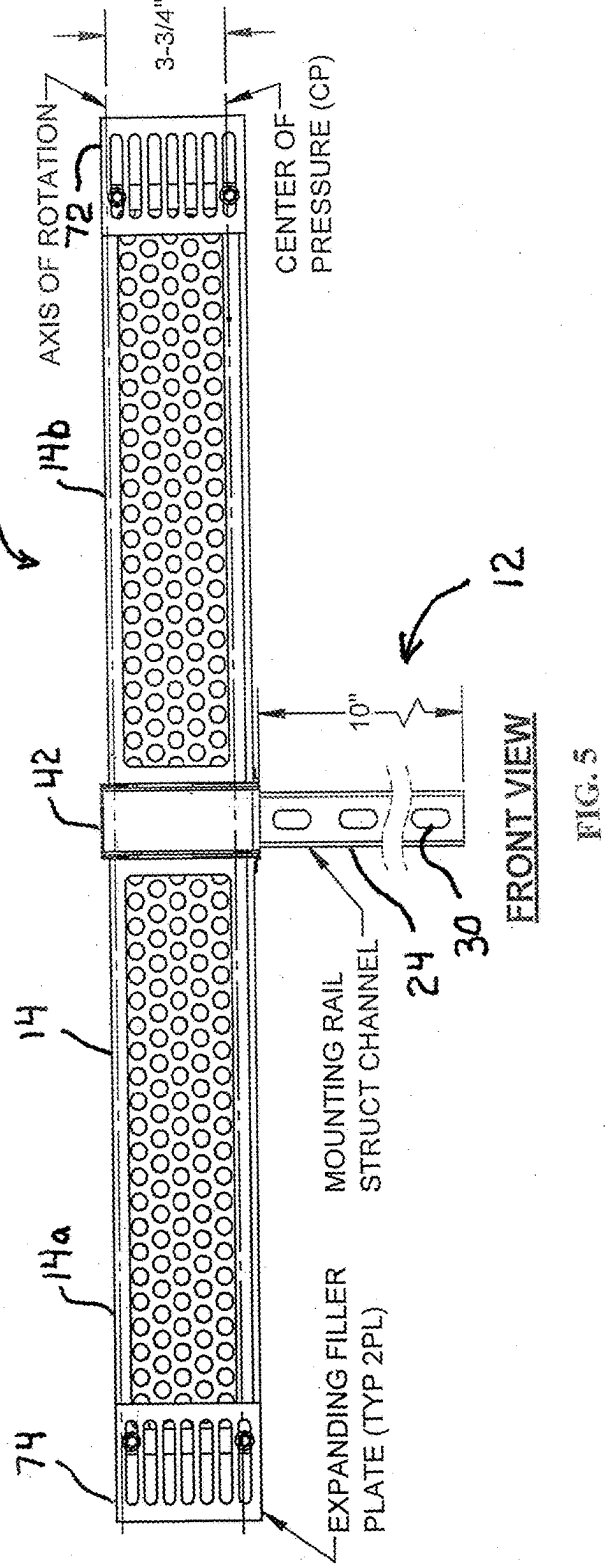

AXIS OF ROTATION
72
3-3/4"
CENTER OF PRESSURE (CP)
14b
10
12
10"
42
14
14a
74
24
30
MOUNTING RAIL
STRUCT CHANNEL
EXPANDING FILLER PLATE (TYP 2PL)
FRONT VIEW

FIG. 5

AUTOMATIC RETRACTABLE SCREEN DEVICE FOR STORM DRAINS HAVING CURBSIDE OPENINGS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Provisional Application Ser. No. 63/621,286, filed on Jan. 16, 2024, and titled "Automatic Retractable Screen Device For Storm Drains Having Curbside Openings", the disclosure of which is incorporated herein by reference and on which priority is hereby claimed.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention generally relates to stormwater drainage systems, and more particularly relates to filtering apparatus used therewith. Even more specifically, the present invention relates to Automatic Retractable Screen (ARS) devices used in conjunction with storm drains formed with curbside openings.

Description of the Prior Art

[0003] Stormwater drainage systems are commonly used in streets, highways, parking lots and other paved surfaces throughout the United States and in foreign countries, especially in urban developments, to remove water accumulating on the surfaces thereof. In many municipalities, curbside storm vaults, or stormwater catch basins, are installed below grade along such paved roadways. Such catch basins are usually formed from concrete and are connected to a conduit that channels the stormwater away from the area to a relatively large water collection reservoir, or to a river, stream or other body of water.

[0004] A typical storm vault **2** which is used extensively in the streets of many municipalities in the United States is shown in FIG. **1** of the drawings. Such storm vaults **2** are usually made from concrete and are rectangular in transverse and longitudinal cross-sections, i.e., the vaults **2** are formed as a rectangular parallelepiped, with opposite front and rear side walls, opposite lateral side walls, a bottom wall and a top cover wall. The walls of the storm vault **2** define an interior cavity **3** which receives stormwater through an elongated, curbside opening, or open-curb stormwater inlet **4**, formed in the front wall of the concrete vault **2**, the vault **2** being positioned in the ground such that the curbside opening **4** is level with the surface of the paved roadway to receive runoff water accumulating on the roadway therethrough and into the internal cavity of the vault **2**, where the water is directed to a channeling conduit forming part of the stormwater drainage system. There is also an access opening **6** formed through the thickness of the cover wall of the storm vault **2**, which access opening **6** is covered by a storm grate or a manhole cover **8**.

[0005] Many municipalities require the placement of a screen or filtering apparatus at the curbside opening **4** that prevents coarse materials and debris, such as leaves, stones, branches and manmade trash, carried by stormwater from entering the storm vault **2** through the curbside opening **4** formed therein and that collects within the internal cavity of the storm vault **2**. Such coarse debris material may clog the drainage conduit connected to the storm vault **4** and may contribute to the pollution of the body of water to which the stormwater from the catch basin (storm vault) **4** collects.

[0006] One type of device used to protect curb inlets **4** from trash and debris that might typically enter an open curb inlet and accumulate in the below-ground storm vault **2** is referred to in the stormwater industry as an Automatic Retractable Screen, or simply, ARS, device. The basic purpose for the ARS device is to provide screen protection at the mouth of curb inlet openings **4** to prevent trash and debris from entering the inlet **4** during the dry season or during light rain events. The roadside trash and debris would then be collected during typical street-sweeping routines. The function of the ARS device is to remain closed until a pre-specified height of water in front of the screen of the device has accumulated sufficiently to force the screen to open, thus allowing the curb inlet **4** to accept the full flow of the storm event for which it was designed. The activation from closed to open, and vice versa, is required to occur by mechanical means only, since no electrical power is provided to the storm vault **2**.

[0007] Some conventional ARS designs incorporate springs or counterbalances to perform this function. The spring-activated ARS device inherently has an increasing force (based on spring rate) that introduces more and more force as the screen opens wider to accept the incoming stormwater. In turn, this increasing spring force will create an undesirable higher head of water upstream entering the curb inlet opening. With regard to the counterbalance ARS devices, they tend to require significantly more space to install, and are believed to be very unreliable.

OBJECTS AND SUMMARY OF THE INVENTION

[0008] It is an object of the present invention to provide an improved ARS device that is fittable at the curbside opening of an existing storm drainage basin.

[0009] It is another object of the present invention to provide an improved ARS device for use with a storm vault having a curbside opening, and which has a screen that is adjustable in width to accommodate curbside openings having various widths.

[0010] Is it a further object of the present invention to provide an improved ARS device having a pivotable screen, and which includes structure allowing for adjustment in the breakaway force required to pivot the screen.

[0011] It is yet a further object of the present invention to provide an improved ARS device for use with a storm vault having a curbside opening, the ARS device including a pivotable screen in which, once the required breakaway force for pivoting the screen is met, the screen substantially freely pivots without restriction and closes automatically under its own weight when the flow of stormwater through the curbside opening subsides.

[0012] In accordance with one form of the present invention, an improved ARS device for use with a storm vault having a curbside opening includes a mounting assembly having a bracket for mounting the ARS device to a side wall of the storm vault adjacent to the curbside opening, a pivotable screen, preferably formed of first and second sections arranged side-by-side, positioned inside the curbside opening of the storm vault near the front of the opening, structure affixed to the mounting assembly and the screen which allows the screen to pivot on the mounting assembly, and a magnetic release mechanism. The magnetic release

mechanism holds the screen in a closed position to block debris carried by stormwater from entering the vault opening until the force of the stormwater exerted on the screen exceeds a predetermined and selectable value at which point the screen is released to pivot to an open position.

[0013] More specifically, and in one form of the present invention, the ARS device uses a sealed rare earth permanent magnet to hold the screen in the closed position until a sufficient head of water in front of the screen provides sufficient force to overcome the magnetic attraction between the magnet and a ferrous strike plate (or a second magnet) positioned in close proximity to the magnet and mechanically coupled to the screen, and allows the screen to break free of the magnetic force and pivot freely (with exception to weight and friction) to an open position. The benefit of such a novel design which employs a magnetic release mechanism over the conventional spring-activated ARS device is that, unlike springs used conventionally, once the screen breaks away from the magnet, it pivots with little restriction.

[0014] In a further preferred version of the ARS device of the present invention, the release mechanism which actuates pivotal movement of the screen incorporates a disc-shaped magnet that is fixed to the end of a socket head bolt. Adjusting the bolt moves the magnet closer or farther from a magnetic or magnetically-attractable strike plate operatively coupled to the hinged screen, thus allowing the installer to fine-tune the required breakaway force per the customer's requirements. Once the proper breakaway force is achieved, the bolt and magnet assembly are permanently locked in place with a locknut and thread lock adhesive.

[0015] These and other objects, features and advantages of the present invention will be apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. **1** is a perspective view of a stormwater drainage basin, or storm vault, for which the automatic retractable screen device of the present invention is adapted to be used.

[0017] FIG. **2** is a front perspective view of an Automatic Retractable Screen (ARS) device formed in accordance with the present invention and shown with the screen thereof in a closed position.

[0018] FIG. **3** is a rear perspective view of the ARS device of the present invention with the screen thereof shown in a closed position.

[0019] FIG. **4** is a front elevational view of the ARS device of the present invention with the screen thereof shown in a closed position.

[0020] FIG. **5** is another front elevational view of the ARS device of the present invention shown with the screen thereof in a closed position.

[0021] FIG. **6** is a top plan view of the ARS device of the present invention shown with the screen thereof in a closed position.

[0022] FIG. **7** is an exploded, front, bottom perspective view of the ARS device of the present invention.

[0023] FIG. **8** is an exploded, front perspective view of the ARS device of the present invention.

[0024] FIG. **9** is an exploded, front perspective view of a portion of the ARS device of the present invention.

[0025] FIG. **10** is a side elevational view of the ARS device of the present invention showing the screen thereof in a closed position.

[0026] FIG. **11** is a front perspective view of the ARS device of the present invention shown with the screen thereof in an open position.

[0027] FIG. **12** is a rear perspective view of the ARS device of the present invention shown with the screen thereof in an open position.

[0028] FIG. **13** is a side view of a portion of the ARS device of the present invention showing the screen thereof in a closed position and further illustrating a preferred form of the magnetic release mechanism of the ARS device.

[0029] FIG. **14** is a partially exploded, perspective view of a portion of the mounting assembly of the ARS device of the present invention.

[0030] FIG. **15** is a longitudinal cross-sectional view of the ARS device of the present invention taken through the center thereof and showing the screen thereof in a closed position, and further illustrating a preferred form of the magnetic release mechanism of the device.

[0031] FIG. **16** is an enlarged rear perspective view of a portion of the ARS device of the present invention and illustrating the preferred form of the magnetic release mechanism of the device.

[0032] FIG. **17** is an exploded perspective view of a portion of the magnetic release mechanism of the ARS device of the present invention.

[0033] FIG. **18** is a side view of the ARS device of the present invention showing the screen thereof in a fully open position.

[0034] FIG. **19** is a front perspective view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault which is shown in transparency, and further showing the screen thereof in a closed position.

[0035] FIG. **20** is a side elevational view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault which is shown in transparency, and further showing the screen thereof in a closed position.

[0036] FIG. **21** is a partially exploded, side elevational view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault, and further showing the screen thereof in a closed position.

[0037] FIG. **22** is a front perspective view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault which is shown in transparency, and further showing the screen of the ARS device in a closed position.

[0038] FIG. **23** is an enlarged, front perspective view of the ARS device of the present invention and a portion of the storm vault shown in FIG. **22**, and illustrating its placement within the curbside opening of the storm vault which is shown in transparency, and further showing the screen of the ARS device in a closed position.

[0039] FIG. **24** is a side elevational view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault, and further showing the screen thereof in a closed position.

[0040] FIG. **25** is a side elevational view of the ARS device of the present invention and illustrating its placement

within the curbside opening of a storm vault which is shown in transparency, and further showing the screen thereof in an open position.

[0041] FIG. **26** is a front perspective view of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault which is shown in transparency, and further showing the screen thereof in an open position.

[0042] FIG. **27** is an enlarged, front perspective view of the ARS device of the present invention and a portion of the storm vault shown in FIG. **26**, taken from a different angle from that of FIG. **26**, and illustrating its placement within the curbside opening of a storm vault which is shown in transparency, and further showing the screen of the ARS device in an open position.

[0043] FIG. **28** is a side elevational view of a portion of the ARS device of the present invention and illustrating its placement within the curbside opening of a storm vault, and which the screen of the ARS device is shown in a fully open position.

[0044] FIG. **29** is a vector force diagram of the forces applied to the screen of the ARS device of the present invention by stormwater contacting the screen.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0045] As can be seen from FIGS. **1-28** of the drawings, an Automatic Retractable Screen (ARS) device **10** constructed in accordance with the present invention, in its most basic form, includes a mounting assembly **12** for mounting the ARS device **10** to a side wall of the storm vault **2** adjacent to the curbside opening **4**, a pivotable screen **14**, preferably formed of first and second sections **14***a*, **14***b* arranged side-by-side, positioned inside the curbside opening **4** of the storm vault **2** near the front **16** of the opening **4**, screen pivot structure **18** affixed to the mounting assembly **12** and the screen **14** which allows the screen **14** to pivot on the mounting assembly **12**, and a magnetic release mechanism **20**. Each of these components will now be described in greater detail.

[0046] As can be seen in FIGS. **2**, **7**, **9** and **11-14**, the mounting assembly **12** of the ARS device **10** of the present invention not only is used to mount the ARS device **10** to a side wall **22** of a storm vault **2** but also allows adjustment of the pivotable screen **14** in depth and angular disposition within the curbside opening **4** of the storm vault **2**. The mounting assembly **12** includes an elongated bracket **24** which is U-shaped in transverse cross section and formed of two spaced apart lateral side walls **26** which extend perpendicularly from a base wall **28** which faces the inner surface of a side wall **22** of the storm vault **2** when affixed thereto. The elongated bracket **24** includes a plurality of spaced apart slots **30** positioned along the length thereof and through the thickness of the base wall **28**. The slots **30** are provided to receive concrete anchors **32** for mounting the elongated bracket **24** in a desired position against the inner surface of a side wall **22** of the storm vault **2** situated below the curbside opening **4**.

[0047] The mounting assembly **12** further includes a pair of support plates **34** which extend perpendicularly from a top end portion of the elongated bracket **24** and which are joined to the lateral side walls **26** of the bracket **24**. Each support plate **34** has formed through the thickness thereof a slot **36** which extends, like the support plates **34** themselves, perpendicularly to the longitudinal length of the elongated bracket **24**. As will be described in greater detail, a machine bolt **38** is received by and passes through the slots **36** formed in the support plates **34**, the threaded end of the shank of which receives a nut **40**. The slots **36** are provided to allow adjustment to the position of the screen **14** within the curbside opening **4** both in lateral disposition and in angular disposition to compensate for differences in the thickness of the vault side wall **22** in which the curbside opening **4** is formed and irregularities in the wall **22** defining the opening **4** and slope of the opening **4**.

[0048] The mounting assembly **12** further includes a housing **42** for holding therewithin the magnetic release mechanism **20**. The housing **42** includes opposite first and second lateral side walls **44**, **46**, spaced apart from and parallel to each other, a generally Z-shaped top covering wall **48** joined to the top edges of the first and second lateral side walls **44**, **46**, a front wall **50** joined to the front edges of the first and second lateral side walls **44**, **46**, and a bottom wall **52** joined perpendicularly to (and preferably integrally formed with) the front wall **50** and joined to the bottom edges of the first and second lateral side walls **44**, **46**. Thus, the housing **42** defines an interior cavity **54** in which at least a portion of the magnetic release mechanism **20** of the ARS device **10** of the present invention is situated.

[0049] Inside the housing **42** is a horizontal mounting plate **56** for mounting the magnetic release mechanism **20**, and a vertical support plate **58**, preferably integrally formed with and perpendicularly disposed to the horizontal mounting plate **56**, each of which extend across the interior cavity **54** of the housing **42** between the first and second lateral side walls **44**, **46**. The machine bolt **38** which is received by the slots **36** formed in the pair of support plates **34** passes through openings **60** formed in the first and second lateral side walls **44**, **46** of the housing **42** and through a bushing **62** disposed between the lateral side walls **44**, **46** so that the housing **42** is mounted between and supported by the support plates **34** of the mounting assembly **12**. The housing **42** is thus adjustable on the support plates **34** relative to the longitudinal length of the elongated bracket **24** and angularly thereto by adjusting its lateral position and angular disposition on the pair of support plates **34** through the positioning of the machine bolt **38** within the slots **36**.

[0050] Each of the first and second lateral side walls **44**, **46** includes a portion **64** that extends upwardly beyond the top covering wall **48** of the housing **42**. As will be described in greater detail, such upwardly extending portions **64** of the first and second lateral side walls **44**, **46** of the housing **42** are used to pivotally support the screen **14** of the ARS device **10** of the present invention. As will be further seen, the housing **42** of the mounting assembly **12** is received at least partially within the curbside opening **4** of the storm vault **2** from the rear **66** thereof.

[0051] As further can be seen in FIGS. **2-8**, **11** and **12**, the screen **14** of the ARS device **10** of the present invention preferably is formed of first and second sections **14***a*, **14***b* arranged co-planarly. More specifically, the first section **14***a* of the screen **14** extends outwardly and perpendicularly to the first lateral side wall **44** of the housing **42** of the mounting assembly **12**, and the second section **14***b* of the screen **14** extends outwardly and perpendicularly from the second lateral side wall **46** of the housing **42**. Each of the first and second screen sections **14***a*, **14***b* is formed with a rectangular frame **68**. A perforated plate **70**, having a mul-

tiplicity of spaced apart holes formed through the thickness thereof, is joined to the rectangular frame **68** of each of the first and second screen sections **14***a*, **14***b*. The plate openings of the first and second sections **14***a*, **14***b* of the screen **14** are provided to allow stormwater to pass therethrough and into the curbside opening **4** of the storm vault **2** when the screen **14** is in a closed position and yet block any debris (leaves, branches, bottles and other manmade articles) carried by the stormwater from passing therethrough and into the interior of the storm vault **2**.

[0052] In a preferred form of the ARS device **10** of the present invention, each of the first and second sections **14***a*, **14***b* of the screen **14** may include an extendable filler plate or grille **72**, **74**, such as shown in FIGS. **2-5**, **7**, **11** and **12** of the drawings. Since the width of the curbside opening **4** may vary from storm vault-to-storm vault, the grilles **72**, **74** may be extended beyond the normal width of the first and second screen sections **14***a*, **14***b* to ensure that the entire width of the curbside opening **4** is protected against debris entering the storm vault **2**. More specifically, the frame **68** of the first screen section **14***a* at the outermost lateral side thereof has adjustably mounted to it a first filler plate or grille **72** having a plurality of spaced apart slots **76** extending transversely therein. The outermost lateral side of the frame **68** includes top and bottom openings **78** formed through the thickness thereof and captively holding rivet nuts **80** which receive machine bolts **82** that also pass through the uppermost and lowermost slots **76** formed in the first grille **72**.

[0053] Similarly, the second screen section **14***b* includes a second filler plate or grille **74**. More specifically, the frame **68** of the second screen section **14***b* at the outermost lateral side thereof has adjustably mounted to it a second filler plate or grille **74** having a plurality of spaced apart slots **76** extending transversely therein. The outermost lateral side of the frame **68** includes top and bottom openings **78**, formed through the thickness thereof and captively holding rivet nuts **80** which receive machine bolts **82** that also pass through the uppermost and lowermost slots **76** formed in the second grille **74**.

[0054] Thus, the first and second grilles **72**, **74** may be extended outwardly from the first and second screen sections **14***a*, **14***b*, respectively, and held in place by tightening the machine bolts **82** so that the perforated screen **14** covers the entire width of the curbside opening **4** of the storm vault **2**.

[0055] The perforated screen **14** is pivotally mounted on the housing **42** of the mounting assembly **12** that holds the magnetic release mechanism **20**. The pivot structure **18** which allows the screen **14** to pivot will now be described.

[0056] As can be seen from FIGS. **3**, **8**, **9**, **11-13** and **16** of the drawings, there is a tubular bar **84** that is welded to the inside surface of the top side of the frames **68** of the first and second screen sections **14***a*, **14***b*. This bar **84** passes through openings **86** formed in the extended upper portions **64** of each of the first and second lateral side walls **44**, **46** of the release mechanism housing **42** and through sleeve bearings or bushings **88** mounted in the openings **86**. Thus, the bar **84** is affixed to and can rotate on the first and second lateral side walls **44**, **46** of the housing **42** so that the first and second screen sections **14***a*, **14***b* joined to the bar **84** can pivot on the housing **42** from a closed position, in which the screen sections **14***a*, **14***b* are substantially parallel with the front wall **50** of the housing **42**, to an inwardly sloped open position, or a perpendicular, fully open position with respect to the front wall **50** of the housing **42**. In this open position of the screen **14**, caused by the force of the stormwater exerted on the screen **14**, the ARS device **10** of the present invention will allow a stormwater surge to freely pass through the curbside opening **4** of the storm vault **2**. When the stormwater surge subsides, the screen **14** by its own weight will return to the closed position covering the curbside opening **4** of the storm vault **2**, allowing stormwater to pass through the perforated screen **14** but not debris carried by the stormwater or in the roadway and collecting at the screen **14**.

[0057] The improved ARS device **10** of the present invention employs a magnetic release mechanism **20** that holds the screen **14** in a closed position to block debris carried by stormwater from entering the vault opening **4** until the stormwater builds up sufficiently to cause the magnetic release mechanism **20** to unlatch and allow the screen **14** to pivot to an open position. Reference should now be had to FIGS. **3**, **8**, **9** and **13-17** which show a preferred form of the magnetic release mechanism **20** of the ARS device **10** of the present invention.

[0058] The horizontal mounting plate **56** within the interior cavity **54** of the housing **42** of the mounting assembly **12** and in which the magnetic release mechanism **20** is situated has an opening **90** formed through the thickness thereof. A threaded rivet nut **92** is mounted in this opening **90** and extends outwardly from the upper side of the mounting plate **56**. A force adjustment screw (which is preferably factory set) in the form of a threaded machine bolt **94** has its threaded shank **96** passing through the rivet nut **92** and the horizontal mounting plate **56** and extending upwardly through the interior cavity **54** of the housing **42** a predetermined distance above the upper side of the mounting plate **56** and towards the top covering wall **48** of the housing **42**. Between the bolt head **98** and the lower side of the horizontal mounting plate **56** is a hexagonal nut **100** threaded onto the shank **96** of the machine bolt **94**, acting as the force adjustment screw. Tightening this hexagonal nut **100** against the bottom side of the horizontal mounting plate **56** will prevent the machine bolt **94** from rotating and will fix the length of the threaded shank **96** thereof which extends above the top side of the horizontal mounting plate **56**.

[0059] Preferably, an encased rare earth Neodymium permanent magnet **102** having a short threaded stud **104** is affixed to the free end of the threaded shank **96** of the machine bolt **94** using a thumb nut or standoff **106** having a female/female thread on each end. An adhesive, such as a permanent (or non-permanent) thread lock glue, is applied to the threaded stud **104** of the encased magnet **102** and to the threaded free end of the bolt **94** prior to their being threaded into the thumb nut **106** to prevent the parts from inadvertently separating. Preferably, the encased Neodymium magnet **102** having a threaded stud **104** is Part No. 3506K64 distributed by McMaster-Carr Supply Company of Douglasville, Georgia.

[0060] Alternative structures to that of the described encased and studded permanent magnet **102** may be suitable for use in the magnetic release mechanism **20** of the ARS device **10** of the present invention. For example, a magnet holder (not shown) preferably in the form of a cylindrical cup may be mounted on the free end of the force adjustment screw **94**. The magnet holder, if such is used, would include a recess formed in the upper surface thereof, which receives and secures in place with an adhesive a permanent rare earth magnet.

[0061] The force adjustment screw (e.g., machine bolt **94**) may be a hex drive threaded screw or have a Phillips or slotted head **98**. An opening **108** may be formed in the bottom wall **52** of the housing **42** and disposed in alignment with the head **98** of the screw **94** to provide access to the screw head **98** using a hex wrench or a screwdriver to adjust the position of the permanent magnet **102** affixed to the free end of the screw **94** within the housing **42**.

[0062] A clevis-like actuating bracket **110** includes a top wall **112** that extends over the permanent magnet **102** and the top covering wall **48** of the housing **42** that holds the magnetic release mechanism **20** therein. The actuating bracket **110** includes two articulated, spaced apart, parallel lateral side walls **114** that extend from the top wall **112** of the actuating bracket **110**. The free ends of the articulated lateral side walls **114** of the actuating bracket **110** are respectively joined to the inside surfaces of the innermost lateral sides of the frames **68** of the first and second screen sections **14***a*, **14***b* which are closest to the housing **42** for the magnetic release mechanism **20**. A strike plate **116** formed of a ferrous or other magnetic material, or even defined by a second magnet, is affixed to the underside of the top wall **112** of the actuating bracket **110** at least in partial alignment with and overlying the permanent magnet **102** of the magnetic release mechanism **20**. Accordingly, the magnetic attraction between the permanent magnet **102** of the magnetic release mechanism **20** and the strike plate **116** of the actuating bracket **110**, through the housing top covering wall **48** that is made from a non-magnetic material, will maintain the screen **14** in a closed position within the curbside opening **4** of the storm vault **2**. However, if sufficient force is exerted on the screen **14** of the ARS device **10** by a stormwater surge that overcomes the magnetic attraction between the permanent magnet **102** and the strike plate **116**, such will cause the strike plate **116** and actuating bracket **110** to lift away from the permanent magnet **102** and the housing **42** so that the screen **14** may pivot inwardly of the curbside opening **4** to allow the surge of stormwater to pass therethrough and into the storm vault **2**. When the force exerted on the screen **14** of the ARS device **10** by the surge of stormwater subsides, the screen **14** by its own weight will pivot towards a closed position, with the strike plate **116** affixed to the top wall **112** of the actuating bracket **110** again residing over the permanent magnet **102** of the magnetic release mechanism **20**. Magnetic attraction between the strike plate **116** and the permanent magnet **102** through the housing top covering wall **48** will latch the screen **14** of the ARS device **10** in the closed position until again being overcome by water pressure caused by a stormwater surge. The ARS device **10** of the present invention with the screen **14** in the open position is shown in FIGS. **11**, **12**, **18** and **25-28** of the drawings. The ARS device **10** of the present invention with the screen **14** in the closed position is shown in FIGS. **2-10**, **13**, **15**, **16** and **19-24** of the drawings.

[0063] The force adjustment screw, which as described previously is preferably in the form of a threaded machine bolt **94**, may be turned clockwise or counterclockwise to adjust the position of the permanent rare earth magnet **102** with respect to the ferrous or magnetic strike plate **116** mounted on the actuating bracket **110** situated above the magnet **102**. By adjusting the relative position of the permanent magnet **102** with respect to the strike plate **116**, and thus the degree of magnetic attraction between the strike plate **116** and the permanent magnet **102**, the amount of force caused by stormwater required to overcome the magnetic attraction to unlatch the screen **14** to pivot from a closed position to an open position may be adjusted. The lock nut **100** on the force adjustment screw **94** may be turned to lock in place the position of the permanent magnet **102** relative to the strike plate **116**.

[0064] FIG. **29** is a diagram relating to a vector force free body analysis of the ARS device **10** of the present invention, and this diagram should be read in conjunction with Table I and Equations 1-5 which are set forth below:

TABLE I

ARS Design Aid.

Plate Height (h)	4.473	in	0.373	ft
Plate Length (b) (see bullet note)	39.625	in	3.302	ft
Water Depth for Activation (W_D)	1.500	in	0.125	ft
Distance to Magnet Force F_1	1.000	in	0.083	ft
Plate Angle from Horizontal (θ)	90.0	deg	1.571	rad
H2O Density (Y)	0.036	lb/in^3	62.4	lb/ft^3
Wet Area on Plate (A_W)	59.44	in^2	0.413	ft^2
Top of Plate to Water Surface (W_S)	2.973	in	0.24779	ft
Height of Water Force Centerline (h_{CL})	0.750	in	0.063	ft
2nd Moment of Inersia (I_{XX})	11.1	in^4	5.4E−04	ft^4
Center of Pressure (Y_{CP})	0.250	in	0.021	in
Pressure @ Centerline Height (P_{CL})	0.027	psi		
Pressure @ Bottom (Pmax)	0.054	psi		

$$F = P * A \quad \text{Equation 1}$$

$$P = p * g * h \quad \text{Equation 2}$$

$$Y = p * g = 62.4 \text{ lb/ft}^3 \quad \text{Equation 3}$$

$$Y_{CP} = \frac{I_{XX} * \text{Sin}(\theta)}{h_{CL} * A_W} \quad \text{Equation 4}$$

$$I_{XX} = \frac{b * h^3}{12} \quad \text{Equation 5}$$

[0065] More specifically, Table I and FIG. **29** describe the vector forces acting on the ARS device **10** of the present invention and show how the proper magnetic force (F_B) is determined for a particular application, with respect to the physical dimensions of the screen **14** of the ARS device **10**, including the screen height, also referred to as the plate height (h), from the pivot point of the screen **14** to its lowermost extent, the overall width of the screen **14**, also referred to as the plate length (b), and the distance of the magnetic strike plate **116** to the magnet force (F_1), as well as the required water depth (W_D) adjacent the screen **14** for activation of the ARS device **10** to unlatch the screen **14** and allow the screen **14** to pivot from a closed position to an open position.

[0066] Stated another way, and with reference to Table I and FIG. **29**, the actual screen height is defined as "h", and the moment arm is the distance from the center of rotation "A" to the force acting at the center of pressure "F_{CP}". The resulting force "F_{CP}" and moment arm are calculated and based on the depth of the water "W_D" pushing on the screen. "$\Sigma M_A=0$" states that the sum of the moments about point "A" (point of rotation) must equal zero just prior to adding sufficient horizontal force at plane F_{CP}, to overcome (break away) the vertical magnetic force acting at distance A-B.

The water depth for activation is preferably provided by the customer who purchases the ARS device **10**.

[0067] Even more specifically, Table I and FIG. **29** are provided as an example to show the vector force free body analysis of an ARS device **10** constructed in accordance with the present invention and having particular dimensions. In this example of the ARS device **10**, the screen or plate height (h) is selected as 4.473 inches, the screen width or plate length (b) is selected as 39.625 inches, the water depth (W_D) for activating the ARS device **10** of the present invention is chosen as 1.500 inches, the distance-to-magnet force (F_1) is 1.000 inches, the screen or plate angle from the horizontal (θ) is 90.0 degrees, and the water (H_2O) density (Y) is 0.036 lb/in^3, the wet area on the screen or plate (A_W) is 59.44 in^2, the top of the screen or plate to the water surface (W_S) is 2.973 inches, the height of the water force centerline (h_{CL}) is 0.75 inches, the second moment of inertia (I_{XX}) is 11.1 in^4, the center of pressure (Y_{CP}) is 0.25 inches, the pressure at the centerline height (P_{CL}) is 0.027 psi and the pressure at the bottom of the screen (Pmax) is 0.054 psi.

[0068] Equations 1 through 3 are standard equations for basic fluid analysis and are well known to one skilled in the art. Equation 4 is used to determine the center of pressure, and Equation 5 is used to determine the second moment of inertia for a rectangle (which is substantially the shape of the screen **14**), and these equations are also well known to one skilled in the art.

[0069] In determining the vector forces on the ARS device **10** of the present invention, the following notes should be observed:

[0070] Use the Combined length of each Plate (i.e., screen **14**) for (b) value. Do not include the fixed center magnet housing **42** in this measurement.

[0071] The magnitude of force at the center of pressure (F_{CP}) is always equal to the centerline force (F_{CL}) but acts at a distance Y_{CP} lower than the centerline force.

[0072] Only forces F_B and F_{CP} are required for the Free Body moment analysis.

[0073] This analysis does not account for the weight or friction of the moving parts; these effects will be determined empirically.

[0074] Reference should now be had to FIG. **24** of the drawings which shows the ARS device **10** of the present invention installed in the curbside opening **4** of a storm vault **2**. It may be seen from FIG. **24** how the ARS device **10** is mounted to the inner surface of the side wall **22** of the storm vault **2** that is adjacent to the curbside opening **4** and how it is positioned within the curbside opening **4**. It can also clearly be seen from FIG. **24** of the drawings that the screen **14** may be positioned near the front **16** or mouth of the curbside opening **4** by adjusting the position of the housing **42** that holds the magnetic release mechanism **20** on the pair of support plates **34** of the mounting assembly **12** and where the machine bolt **38** resides in the slot **36** of each support plate **34**. Preferably, the length of the slots **36** provide a lateral adjustment of the position of the screen **14** within the curbside opening **4** of about one and a half (1½) inches, which should be sufficient for most anticipated applications. This lateral adjustment is depicted in FIG. **10** of the drawings. Furthermore, and as mentioned previously, the housing **42** for the magnetic release mechanism **20** may be rotated on the pair of support plates **34** of the mounting assembly **12** by loosening the machine bolt **38** received in the slots **36** of the support plates **34**. This adjustment is provided to the ARS device **10** of the present invention to accommodate curbside openings **4** which are not necessarily horizontal. In a preferred form of the ARS device **10** of the present invention, and as shown in FIG. **10**, a rotational adjustment of +30 degrees to −30 degrees of the housing **42** and screen **14** with respect to the support plates **34** of the mounting assembly **12** is provided, as this amount of rotational adjustment should be sufficient to compensate for non-horizontal, or sloping, curbside openings **4** in most applications.

[0075] As also shown in FIG. **24**, the height of the screen **14** of the ARS device **10** of the present invention is preferably chosen to be less than the height of the curbside opening **4** in which it is placed so as to leave a bypass gap **118** between the screen **14** and the interior vault wall **22** defining the top of the curbside opening **4**. This bypass gap **118** is provided in the event a stormwater surge is so great that additional unrestricted flow of water into the storm vault **2** through the bypass gap **118** is required or if the screen **14** becomes clogged.

[0076] The Automatic Retractable Screen (ARS) device **10** of the present invention employs a magnetic release mechanism **20** which allows the screen **14** to pivot with little restriction once the magnetic attraction between the magnet **102** and the strike plate **116** is overcome. Such a design overcomes the disadvantages inherent with conventional spring-actuated ARS devices which introduce more and more force as the screen opens wider to accept the incoming stormwater. This increasing spring force of conventional ARS devices creates an undesirable higher head of water upstream entering the curb inlet **4**. Additionally, the ARS device **10** of the present invention is simple in construction and includes designed-in adjustability features to help account for irregularities in the curb inlets **4** of typical storm vaults **2**. Furthermore, the expanding filler plates or grilles **72**, **74** adjustably mounted on the first and second sections **14***a*, **14***b* of the screen **14** are provided to account for variations in curb inlet mouth widths.

[0077] The ARS device **10** of the present invention is preferably constructed of non-magnetic 300 series stainless steel with the exception of the permanent magnet **102** (preferably, a Neodymium rare earth magnet) and the strike plate **116** (preferably, formed from a nickel-coated, low carbon steel), which is magnetic.

[0078] The automatic retractable screen device **10** of the present invention will now be further described.

[0079] In a first embodiment of the present invention, an automatic retractable screen device **10** for use with a storm vault **2** is disclosed herein. The storm vault **2** defines an interior cavity **3** and has a side wall **22** and a vault opening **4** formed in the side wall **22**, the vault opening **4** allowing stormwater to flow therethrough and into the interior cavity **3** of the storm vault **2**. The automatic retractable screen device **10** in accordance with the first embodiment of the present invention preferably comprises a mounting assembly **12** for mounting the automatic retractable screen device **10** to the side wall **22** of the storm vault **2** and in close proximity to the vault opening **4**; a pivotable screen **14** positionable to be in alignment with the vault opening **4** of the storm vault **2** when the automatic retractable screen device **10** is mounted within the interior cavity **3** of the storm vault **2**; structure affixed to the mounting assembly **12** and the screen **14** and configured to allow the screen **14** to pivot on the mounting assembly **12** between a closed position and an open position, the open position being different from the

closed position; and a magnetic release mechanism **20** operatively coupled to the screen **14**, wherein the magnetic release mechanism **20** holds the screen **14** in the closed position to block debris carried by the stormwater from entering the vault opening **4** until a force of the stormwater exerted on the screen **14** exceeds a predetermined value at which point the screen **14** is released to pivot to the open position.

[0080] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the mounting assembly **12** may include a bracket **24** configured to be affixable to the side wall **22** of the storm vault **2**.

[0081] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the mounting assembly **12** may include at least one support plate **34**, the at least one support plate **34** having a slot **36** formed through the thickness thereof, and a fastener **38** received by the slot **36** and adjustably positionable along a length of the slot **36**. The structure configured to allow the screen **14** to pivot between the closed position and the open position is operatively coupled to the fastener **38** received by the slot **36** and is adjustable in position relative to the at least one support plate **34** so that the screen **14** is adjustably positionable relative to the vault opening **4** when the automatic retractable screen device **10** is mounted within the interior cavity **3** of the storm vault **2**.

[0082] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the magnetic release mechanism **20** may include a permanent magnet **102**, a strike plate **116** having a structure which is magnetically attractable to the permanent magnet **102**, and a bracket **110** interconnecting the strike plate **116** and the screen **14**, the strike plate **116** being disposed in at least partial alignment with and in close proximity to the permanent magnet **102** when the screen **14** is in the closed position, the permanent magnet **102** and the strike plate **116** being spaced apart by a separation distance when the screen **14** is in at least one of the closed position and the open position.

[0083] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the magnetic release mechanism **20** may include structure for adjusting the separation distance between the permanent magnet **102** and the strike plate **116**.

[0084] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the mounting assembly **12** may include a mounting plate **56**, and the magnetic release mechanism **20** may further include an elongated member **94**, the elongated member **94** being adjustably mounted on the mounting plate **56**, the elongated member **94** having an axial end on which is mounted the permanent magnet **102**, the axial end of the elongated member **94** and the permanent magnet **102** mounted thereon extending outwardly from the mounting plate **56** an adjustable distance thereby causing the separation distance between the permanent magnet **102** and the strike plate **116** to be adjusted.

[0085] In the first embodiment of the automatic retractable screen device **10** of the present invention described above, the screen **14** may include opposite lateral edges, and the automatic retractable screen device **10** may further comprise screen extension members **72**, **74**, each screen extension member **72**, **74** being adjustably mounted on a respective opposite lateral edge of the screen **14** to extend outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen **14** a desired distance in order to provide a combination of the screen **14** and the screen extension members **72**, **74** with an adjustable overall width.

[0086] In a second embodiment of the present invention, an automatic retractable screen device **10** for use with a storm vault **2** is disclosed herein. As mentioned previously, the storm vault **2** defines an interior cavity **3** and has a side wall **22** and a vault opening **4** formed in the side wall **22**, the vault opening **4** allowing stormwater to flow therethrough and into the interior cavity **3** of the storm vault **2**. The automatic retractable screen device **10** in accordance with the second embodiment of the present invention preferably comprises a bracket **24** configured to be affixable to the side wall **22** of the storm vault **2**; at least one support plate **34** affixed to the bracket **24** and extending therefrom, the at least one support plate **34** having a slot **36** formed through the thickness thereof, and a fastener **38** received by the slot **36** and adjustably positionable along a length of the slot **36**; a housing **42**, the housing **42** being mounted to the at least one support plate **34** by the fastener **38** and being adjustable on the at least one support plate **34** by adjustment of the position of the fastener **38** in the slot **36** of the at least one support plate **34**; a pivotable screen **14** positionable to be in alignment with the vault opening **4** of the storm vault **2** when the automatic retractable screen device **10** is mounted within the interior cavity **3** of the storm vault **2**; a pivot rod **84** operatively coupled to the housing **42** and the screen **14** and configured to allow the screen **14** to pivot on the housing **42** between a closed position and an open position, the open position being different from the closed position; and a magnetic release mechanism **20** operatively coupled to the screen **14**, wherein the magnetic release mechanism **20** holds the screen **14** in the closed position to block debris carried by the stormwater from entering the vault opening **4** until a force of the stormwater exerted on the screen **14** exceeds a predetermined value at which point the screen **14** is released to pivot to the open position.

[0087] In the second embodiment of the automatic retractable screen device **10** of the present invention described above, the housing **42** may define an interior cavity **54**, and the magnetic release mechanism **20** may at least partially reside in the interior cavity **54** of the housing **42**. Furthermore, the housing **42** may include a mounting plate **56** disposed within the interior cavity **54** thereof, and the magnetic release mechanism **20** may include an elongated member **94** mounted on the mounting plate **56**, the elongated member **94** having an axial end, a permanent magnet **102** mounted on the axial end of the elongated member **94**, a strike plate **116** having a structure which is magnetically attractable to the permanent magnet **102**, and a bracket **110** interconnecting the strike plate **116** and the screen **14**, the strike plate **116** being disposed in at least partial alignment with and in close proximity to the permanent magnet **102** when the screen **14** is in the closed position, the permanent magnet **102** and the strike plate **116** being spaced apart by a separation distance when the screen **14** is in at least one of the closed position and the open position.

[0088] In the second embodiment of the automatic retractable screen device **10** of the present invention described above, the elongated member **94** may be adjustably mounted on the mounting plate **56**, the axial end of the elongated member **94** and the permanent magnet **102** mounted thereon extending outwardly from the mounting plate **56** an adjust-

able distance thereby causing the separation distance between the permanent magnet **102** and the strike plate **116** to be adjusted.

[0089] In the second embodiment of the automatic retractable screen device **10** of the present invention described above, the screen **14** may include opposite lateral edges, and the automatic retractable screen device **10** may further comprise screen extension members **72**, **74**, each screen extension member **72**, **74** being adjustably mounted on a respective opposite lateral edge of the screen **14** to extend outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen **14** a desired distance in order to provide a combination of the screen **14** and the screen extension members **72**, **74** with an adjustable overall width.

[0090] In a third embodiment of the present invention, an automatic retractable screen device **10** for use with a storm vault **2** is disclosed herein. As mentioned previously, the storm vault **2** defines an interior cavity **3** and has a side wall **22** and a vault opening **4** formed in the side wall **22**, the vault opening **4** allowing stormwater to flow therethrough and into the interior cavity **3** of the storm vault **2**. The automatic retractable screen device **10** in accordance with the third embodiment of the present invention preferably comprises a bracket **24** configured to be affixable to the side wall **22** of the storm vault **2**; at least one support plate **34** affixed to the bracket **24** and extending therefrom, the at least one support plate **34** having a slot **36** formed through the thickness thereof, and a fastener **38** received by the slot **36** and adjustably positionable along a length of the slot **36**; a housing **42** defining an interior cavity **54**, the housing **42** being mounted to the at least one support plate **34** by the fastener **38** and being adjustable angularly on the at least one support plate **34** and in relative position with respect to the at least one support plate **34** by adjustment of the position of the fastener **38** in the slot **36** of the at least one support plate **34**; a pivotable screen **14** positionable to be in alignment with the vault opening **4** of the storm vault **2** when the automatic retractable screen device **10** is mounted within the interior cavity **3** of the storm vault **2**, the screen **14** being adjustable in distance relative to the vault opening **4** and in angular disposition relative thereto by adjustment of the housing **42** on the at least one support plate **34** when the automatic retractable screen device **10** is mounted within the interior cavity **3** of the storm vault **2**, the screen **14** having an overall weight; a pivot rod **84** operatively coupled to the housing **42** and the screen **14** and configured to allow the screen **14** to pivot on the housing **42** between a closed position and an open position, the open position being different from the closed position; and a magnetic release mechanism **20** operatively coupled to the screen **14** and at least partially residing within the interior cavity **54** of the housing **42**, wherein the magnetic release mechanism **20** holds the screen **14** in the closed position to block debris carried by the stormwater from entering the vault opening **4** until a force of the stormwater exerted on the screen **14** exceeds a predetermined and selectable value at which point the screen **14** is released to pivot to the open position. Preferably, the housing **42** includes a mounting plate **56** disposed within the interior cavity **54** thereof. The magnetic release mechanism **20** preferably includes an elongated member **94** mounted on the mounting plate **56**, the elongated member **94** having an axial end, a permanent magnet **102** mounted on the axial end of the elongated member **94**, a strike plate **116** having a structure which is magnetically attractable to the permanent magnet **102**, and a bracket **110** interconnecting the strike plate **116** and the screen **14**, the strike plate **116** being disposed in at least partial alignment with and in close proximity to the permanent magnet **102** when the screen **14** is in the closed position such that the strike plate **116** and the permanent magnet **102** are in magnetic attraction to each other when the screen **14** is in the closed position to maintain the screen **14** in the closed position until the predetermined and selectable value of the stormwater force is exerted on the screen **14**, the permanent magnet **102** and the strike plate **116** being spaced apart by a separation distance when the screen **14** is in at least one of the closed position and the open position. When the stormwater force exerted on the screen **14** no longer exceeds the predetermined and selectable value, the screen **14** will pivot under the overall weight thereof to the closed position.

[0091] In the third embodiment of the automatic retractable screen device **10** of the present invention described above, the screen **14** may include opposite lateral edges, and the automatic retractable screen device **10** may further comprise screen extension members **72**, **74**, each screen extension member **72**, **74** being adjustably mounted on a respective opposite lateral edge of the screen **14** to extend outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen **14** a desired distance in order to provide a combination of the screen **14** and the screen extension members **72**, **74** with an adjustable overall width.

[0092] In the third embodiment of the automatic retractable screen device **10** of the present invention described above, the screen **14** may include a first screen section **14***a* and a second screen section **14***b*, the housing **42** being interposed between the first screen section **14***a* and the second screen section **14***b*.

[0093] Although illustrative embodiments of the present invention have been described herein with reference to the accompanying drawings, it is to be understood that the invention is not limited to those precise embodiments, and that various other changes and modifications may be effected therein by one skilled in the art without departing from the scope or spirit of the invention.

What is claimed is:

1. An automatic retractable screen device for use with a storm vault, the storm vault defining an interior cavity and having a side wall and a vault opening formed in the side wall, the vault opening allowing stormwater to flow therethrough and into the interior cavity of the storm vault, the automatic retractable screen device comprising:

a mounting assembly for mounting the automatic retractable screen device to the side wall of the storm vault and in close proximity to the vault opening;

a pivotable screen positionable to be in alignment with the vault opening of the storm vault when the automatic retractable screen device is mounted within the interior cavity of the storm vault;

structure affixed to the mounting assembly and the screen and configured to allow the screen to pivot on the mounting assembly between a closed position and an open position, the open position being different from the closed position; and

a magnetic release mechanism operatively coupled to the screen, wherein the magnetic release mechanism holds the screen in the closed position to block debris carried

by the stormwater from entering the vault opening until a force of the stormwater exerted on the screen exceeds a predetermined value at which point the screen is released to pivot to the open position.

2. An automatic retractable screen device as defined by claim **1**, wherein the mounting assembly includes a bracket configured to be affixable to the side wall of the storm vault.

3. An automatic retractable screen device as defined by claim **1**, wherein the mounting assembly includes at least one support plate, the at least one support plate having a slot formed through the thickness thereof, and a fastener received by the slot and adjustably positionable along a length of the slot; and

wherein the structure configured to allow the screen to pivot between the closed position and the open position is operatively coupled to the fastener received by the slot and is adjustable in position relative to the at least one support plate so that the screen is adjustably positionable relative to the vault opening when the automatic retractable screen device is mounted within the interior cavity of the storm vault.

4. An automatic retractable screen device as defined by claim **1**, wherein the magnetic release mechanism includes a permanent magnet, a strike plate having a structure which is magnetically attractable to the permanent magnet, and a bracket interconnecting the strike plate and the screen, the strike plate being disposed in at least partial alignment with and in close proximity to the permanent magnet when the screen is in the closed position, the permanent magnet and the strike plate being spaced apart by a separation distance when the screen is in at least one of the closed position and the open position.

5. An automatic retractable screen device as defined by claim **4**, wherein the magnetic release mechanism includes structure for adjusting the separation distance between the permanent magnet and the strike plate.

6. An automatic retractable screen device as defined by claim **4**, wherein the mounting assembly includes a mounting plate; and

wherein the magnetic release mechanism further includes an elongated member, the elongated member being adjustably mounted on the mounting plate, the elongated member having an axial end on which is mounted the permanent magnet, the axial end of the elongated member and the permanent magnet mounted thereon extending outwardly from the mounting plate an adjustable distance thereby causing the separation distance between the permanent magnet and the strike plate to be adjusted.

7. An automatic retractable screen device as defined by claim **1**, wherein the screen includes opposite lateral edges; and

wherein the automatic retractable screen device further comprises screen extension members, each screen extension member being adjustably mounted on a respective opposite lateral edge of the screen to extend outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen a desired distance in order to provide a combination of the screen and the screen extension members with an adjustable overall width.

8. An automatic retractable screen device for use with a storm vault, the storm vault defining an interior cavity and having a side wall and a vault opening formed in the side wall, the vault opening allowing stormwater to flow therethrough and into the interior cavity of the storm vault, the automatic retractable screen device comprising:

a bracket configured to be affixable to the side wall of the storm vault;

at least one support plate affixed to the bracket and extending therefrom, the at least one support plate having a slot formed through the thickness thereof, and a fastener received by the slot and adjustably positionable along a length of the slot;

a housing, the housing being mounted to the at least one support plate by the fastener and being adjustable on the at least one support plate by adjustment of the position of the fastener in the slot of the at least one support plate;

a pivotable screen positionable to be in alignment with the vault opening of the storm vault when the automatic retractable screen device is mounted within the interior cavity of the storm vault;

a pivot rod operatively coupled to the housing and the screen and configured to allow the screen to pivot on the housing between a closed position and an open position, the open position being different from the closed position; and

a magnetic release mechanism operatively coupled to the screen, wherein the magnetic release mechanism holds the screen in the closed position to block debris carried by the stormwater from entering the vault opening until a force of the stormwater exerted on the screen exceeds a predetermined value at which point the screen is released to pivot to the open position.

9. An automatic retractable screen device as defined by claim **8**, wherein the housing defines an interior cavity;

wherein the magnetic release mechanism at least partially resides in the interior cavity of the housing;

wherein the housing includes a mounting plate disposed within the interior cavity thereof; and

wherein the magnetic release mechanism includes an elongated member mounted on the mounting plate, the elongated member having an axial end, a permanent magnet mounted on the axial end of the elongated member, a strike plate having a structure which is magnetically attractable to the permanent magnet, and a bracket interconnecting the strike plate and the screen, the strike plate being disposed in at least partial alignment with and in close proximity to the permanent magnet when the screen is in the closed position, the permanent magnet and the strike plate being spaced apart by a separation distance when the screen is in at least one of the closed position and the open position.

10. An automatic retractable screen device as defined by claim **9**, wherein the elongated member is adjustably mounted on the mounting plate, the axial end of the elongated member and the permanent magnet mounted thereon extending outwardly from the mounting plate an adjustable distance thereby causing the separation distance between the permanent magnet and the strike plate to be adjusted.

11. An automatic retractable screen device as defined by claim **8**, wherein the screen includes opposite lateral edges; and

wherein the automatic retractable screen device further comprises screen extension members, each screen extension member being adjustably mounted on a respective opposite lateral edge of the screen to extend

outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen a desired distance in order to provide a combination of the screen and the screen extension members with an adjustable overall width.

12. An automatic retractable screen device for use with a storm vault, the storm vault defining an interior cavity and having a side wall and a vault opening formed in the side wall, the vault opening allowing stormwater to flow therethrough and into the interior cavity of the storm vault, the automatic retractable screen device comprising:

a bracket configured to be affixable to the side wall of the storm vault;

at least one support plate affixed to the bracket and extending therefrom, the at least one support plate having a slot formed through the thickness thereof, and a fastener received by the slot and adjustably positionable along a length of the slot;

a housing defining an interior cavity, the housing being mounted to the at least one support plate by the fastener and being adjustable angularly on the at least one support plate and in relative position with respect to the at least one support plate by adjustment of the position of the fastener in the slot of the at least one support plate;

a pivotable screen positionable to be in alignment with the vault opening of the storm vault when the automatic retractable screen device is mounted within the interior cavity of the storm vault, the screen being adjustable in distance relative to the vault opening and in angular disposition relative thereto by adjustment of the housing on the at least one support plate when the automatic retractable screen device is mounted within the interior cavity of the storm vault, the screen having an overall weight;

a pivot rod operatively coupled to the housing and the screen and configured to allow the screen to pivot on the housing between a closed position and an open position, the open position being different from the closed position; and

a magnetic release mechanism operatively coupled to the screen and at least partially residing within the interior cavity of the housing, wherein the magnetic release mechanism holds the screen in the closed position to block debris carried by the stormwater from entering the vault opening until a force of the stormwater exerted on the screen exceeds a predetermined and selectable value at which point the screen is released to pivot to the open position;

wherein the housing includes a mounting plate disposed within the interior cavity thereof;

wherein the magnetic release mechanism includes an elongated member mounted on the mounting plate, the elongated member having an axial end, a permanent magnet mounted on the axial end of the elongated member, a strike plate having a structure which is magnetically attractable to the permanent magnet, and a bracket interconnecting the strike plate and the screen, the strike plate being disposed in at least partial alignment with and in close proximity to the permanent magnet when the screen is in the closed position such that the strike plate and the permanent magnet are in magnetic attraction to each other when the screen is in the closed position to maintain the screen in the closed position until the predetermined and selectable value of the stormwater force is exerted on the screen, the permanent magnet and the strike plate being spaced apart by a separation distance when the screen is in at least one of the closed position and the open position; and

wherein, when the stormwater force exerted on the screen no longer exceeds the predetermined and selectable value, the screen will pivot under the overall weight thereof to the closed position.

13. An automatic retractable screen device as defined by claim **12**, wherein the screen includes opposite lateral edges; and

wherein the automatic retractable screen device further comprises screen extension members, each screen extension member being adjustably mounted on a respective opposite lateral edge of the screen to extend outwardly from or adjust inwardly towards the respective opposite lateral edge of the screen a desired distance in order to provide a combination of the screen and the screen extension members with an adjustable overall width.

14. An automatic retractable screen device as defined by claim **12**, wherein the screen includes a first screen section and a second screen section, the housing being interposed between the first screen section and the second screen section.

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